

1.1. Definition of Economics and principles of Economics.

Definition of Economics:→

Economics is the social science that studies how individuals, businesses, and governments allocate limited resources to satisfy unlimited human wants.

According to "Lionel Robbins"

"Economics is the science which studies human behavior as a relationship between ends and scarce means which have alternative uses."

Branches of Economics:→

1> Microeconomics: studies individual units such as a consumer, firm, or market (e.g. demand, supply, pricing.)

2> Macroeconomics: studies the overall economy (e.g. national income, inflation, employment, fiscal policy)

Basic principles of Economics:

1> Scarcity: Resources are limited, and choices must be made

2> Opportunity cost: the cost of the next best alternative forgone

3> Marginal Analysis: Decisions are made by comparing marginal cost and marginal benefit.

4> Equilibrium: Balance between demand and supply.

5> Efficiency and productivity: optimal use of resources for maximum output

6) Incentives matter &

people respond to rewards and penalties.

Conclusion: Economics helps in efficient decision-making to achieve maximum satisfaction with minimum resources.

1.2 Introduction to demand, supply, and utility.

1.1 Demand

Definition:

Demand refers to the quantity of a good or service that consumers are willing and able to buy at different prices in a given time period.

Law of demand:

"When the price of a commodity falls, the quantity demanded rises, and vice-versa, other things remaining the same."

Factors affecting demand: price, income, taste, substitute, population etc

2.7 SUPPLY

Definition:

Supply is the quantity of goods that producers are willing to sell at different prices during a given period.

Law of Supply:

"When the price of a commodity rises, the quantity supplied also rises, and vice-versa, other things remaining constant."

Determinants: production cost, technology, number of sellers etc.

3.7 UTILITY:

Definition:

Utility is the satisfaction derived from consuming a good or services.

1.3. DEFINITION OF ENGINEERING ECONOMICS, PRINCIPLES OF ENGINEERING ECONOMICS, AND ITS APPLICATION

Definition:

Engineering Economics is the branch of economics that applies economic principles to engineering problems, to make decisions that result in the best use of resources.

Example :

If a new machine costs Rs. 20,000 more but saves Rs. 5,000 per year, analyze if the extra Rs. 20,000 is justified by the extra saving.

3> principle of minimum cost (Economy) :-
point:

- Choose the alternative that achieves the desired objective with the lowest total cost.
- Reduces waste and ensures efficient use of resources.
- Quality should not be compromised while minimizing cost.

Example :

Between two materials giving the same strength, the one costing less per unit is selected.

4> principle of comparative alternatives:-

point:

- Compare all feasible options before making a decision.
- Consider cost, benefit, performance, and future value of each option.
- The best alternative gives the maximum return for the least investment.

Example :->

Compare diesel, solar, and electric systems for power supply - select the one with best cost-benefit ratio over its lifetime.

5. > Principle of Risk and Uncertainty :
points:

- Every engineering project faces risks like inflation, failure, or market change.
- Future costs and benefits are uncertain, so risks must be analyzed.
- Use tools like probability or sensitivity analysis.

Example :

Before starting a hydro project, engineers assess risk of rainfall variation or equipment breakdown.

6. > Principle of Equivalent Worth
points:

- All alternatives must be compared on the same time basis - present worth (PW), annual worth (AW), or future worth (FW).
- Convert all cash flows to a common base for fair comparison.

Example:

If machine A and machine B have different costs and lives, compare both using present worth to find which is more economical.

7.7 principle of Sunk cost and opportunity cost.

point:

- Sunk cost: past cost that cannot be recovered
 - ignore it in new decisions.
- opportunity cost: value of the best alternative forgone when a choice is made.

Example:

1. money already spent on old machine of Rs.

Sunk cost

If that machine could be sold for Rs. 50,000, that amount is the opportunity cost of keeping it.

Elasticity of Demand

Definition:

Elasticity of demand measures how much the quantity demanded of a good changes when there is a change in its determining factors such as price, income, or price of related goods.

Formula (General Form):

$$E = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in determining factor}}$$

Types of Demand Elasticity.

1. Price Elasticity of Demand ✓
2. Income Elasticity of Demand. ✓
3. Cross Elasticity of Demand. ✓

1. Price Elasticity of Demand

Definition:

Price Elasticity of Demand measures the responsiveness of quantity demanded to a change in the price of the same good.

$$E_P = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}}$$

• Interpretations:

- If $E_P > 1$: Demand is Elastic $\rightarrow$ Quantity changes more than price
- If $E_P < 1$: Demand is Inelastic $\rightarrow$ Quantity changes less than price

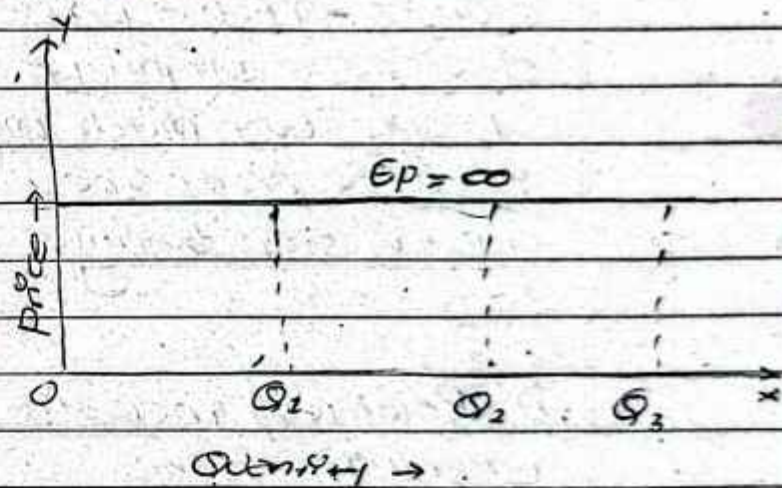
- If $E_p = 1$: unit elastic $\rightarrow$ quantity changes equally as price.

ii) Perfectly Elastic Demand ($E_p = \infty$):

meaning:

Even a very small change in price causes infinite change in demand (buyers will buy only at one price).

Graph: Horizontal demand curve.



iii) Perfectly Inelastic Demand ($E_p = 0$):

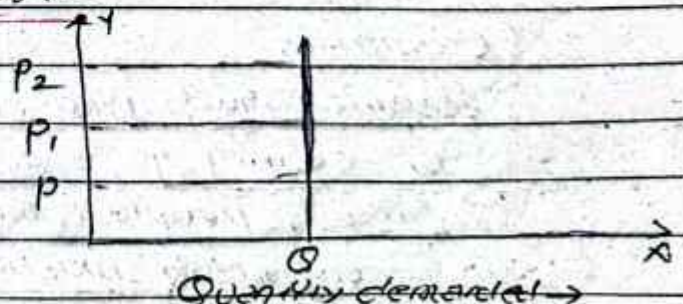
meaning:

change in price does not affect quantity demanded at all.

Graph: Vertical demand curve

Example: Life saving medicines like insulin or heart drugs - even

if price doubles, patients still get the same amount.



(iii) Relatively Elastic Demand ($E_p > 1$)

meaning:

a small change in price causes a larger change in quantity demanded.

Graph: Flatter demand curve ✓

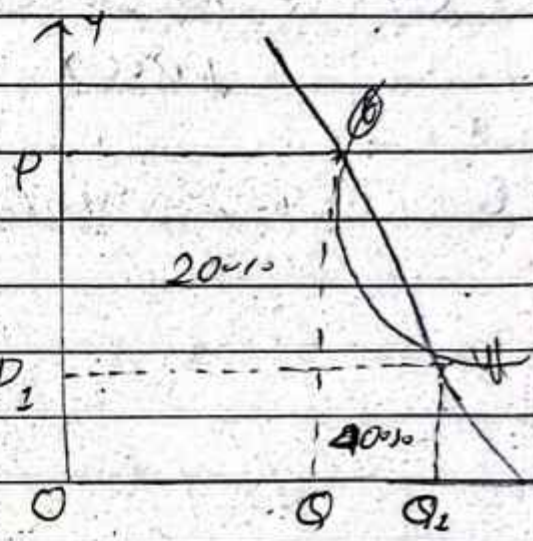
Example:

Luxury goods like

smart phones or branded clothes - if the price decreases slightly,

people buy much more, P_1

if price rises, they quickly stop buying.



(iv) Relatively Inelastic Demand ($E_p < 1$): ✓

meaning:

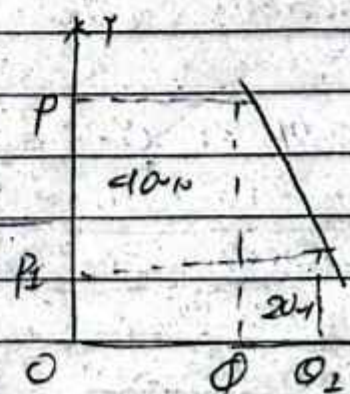
a change in price leads to a smaller change in quantity demanded.

Graph: Steeper demand curve ✓

Example:

Essential goods like salt, rice,

or cooking gas - even if price rises, people cannot reduce consumption much.

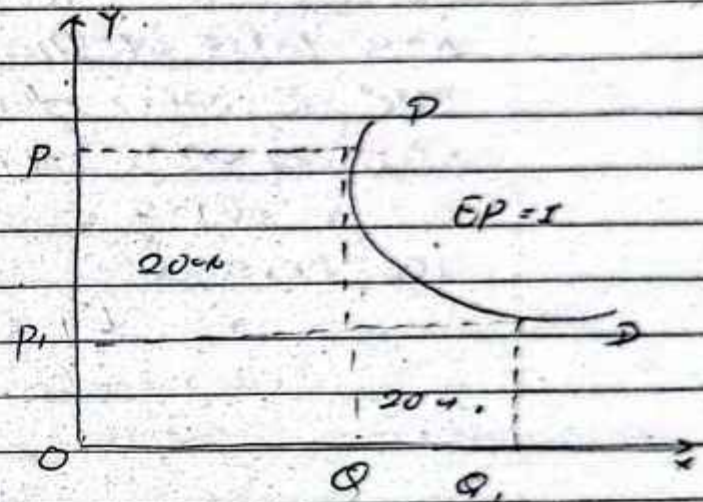


② Unitary Elastic Demand ($E_p = 1$):

Meaning:

percentage change in quantity demanded equals percentage change in price.

Graph: Rectangular hyperbola curve.



Example: Let us suppose that price of a commodity x decreases from Rs 10 per unit to 8 per unit and quantity demanded of x increases from 50 units to 60 units then find out price elasticity of demand.

Given,

Old price of product (P) = Rs 10.

New price of product (P_1) = Rs 8

Old quantity demanded (Q) = Rs 50

New quantity demanded (Q_1) = Rs 60

$$E_p = ?$$

(we have

$$\begin{aligned}
 E_p &= - \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} \\
 &= - \frac{(Q_1 - Q)}{(P_1 - P)} \times \frac{10}{50} \\
 &= - \frac{10}{8 - 10} \times \frac{10}{50} \\
 &= 1
 \end{aligned}$$

2. Income Elasticity of Demand

Definition:

Income Elasticity of Demand measures the responsiveness of the quantity demanded of a good to a change in consumer income:

$$YED = \frac{\text{change in quantity demanded}}{\text{change in income}}$$

Types of Income Elasticity of Demand:

A. Positive Income Elasticity ($E_Y > 0$)

$\Rightarrow$ Goods for which demand increases as income increases.

Normal Goods

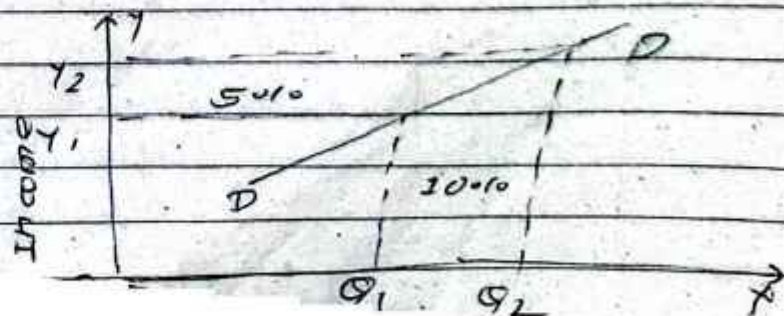
1. Greater than 1 (Luxury Goods)

- $E_Y > 1 \rightarrow$ demand rises more than proportionally with income.

• Example:

- $\hookrightarrow$ Luxury cars (Ferrari, BMW)
- $\hookrightarrow$ High-end smartphones (iPhone, Samsung Galaxy Ultra)
- $\hookrightarrow$ Foreign holidays
- Expansion: If income rises by 10%, demand rises by more than 10%.

Graphically:



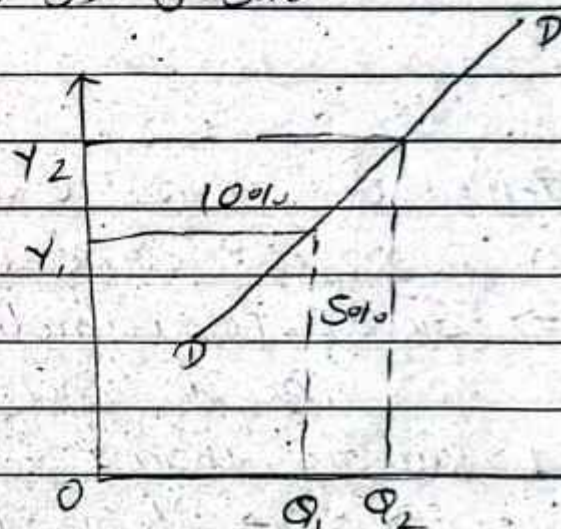
2) Less than 1 (Necessaries)

- $EY < 1 \rightarrow$ demand rises but less than proportionally with income.

• Example:

- Rice, milk, electricity, basic clothing

- Explanation: If income rises by 10% demand increases by 3-5%



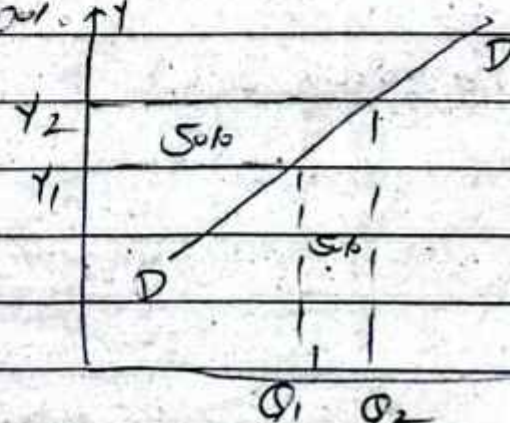
3) Equal to 1 (Luxuries)

- $EY = 1 \rightarrow$ Demand rises exactly proportionally with income.

• Example:

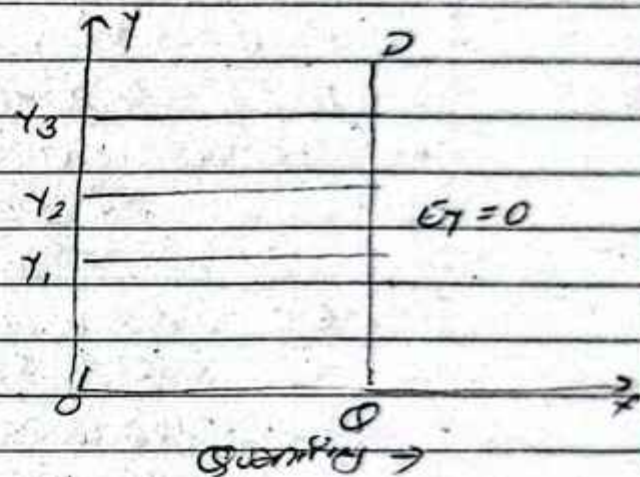
- Average restaurant dining or branded clothing

- Explanation: If income rise by 10% demand also rises by 10%.



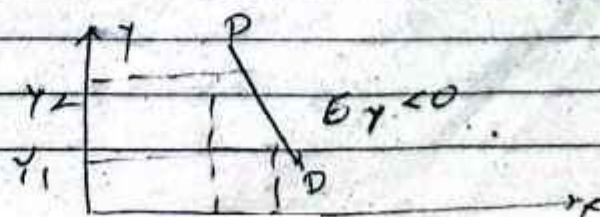
B) Zero income elasticity ($E_Y = 0$)

- Demand does not change with income changes
- Example:-
 - Salt, matches, basic medicines
- Explanation: Even if income increases, consumption remains nearly the same because these goods are essential in fixed quantities.



C) Negative income elasticity ($E_Y < 0$)

- Demand falls as income rises $\rightarrow$ inferior goods
- Example:-
 - public transport (buses, rickshaws) for higher-income people.
 - instant noodles or cheap packaged food.
- Explanation: As income rises, people prefer superior alternatives like private cars, restaurants, or branded foods.



Example 8)

When customer's income rises from Rs. 100 to Rs. 102, his demand for goods X increases from 25 units per week to 30 units per week. then find out income elasticity of demand for good X.

Solution:

$$\text{Initial Income } (Y_1) = 100$$

$$\text{New Income } (Y_2) = 102$$

$$\text{Initial demand } (Q_1) = 25$$

$$\text{New demand } (Q_2) = 30$$

$$E_Y = ?$$

$$E_Y = \frac{\Delta Q}{Q} \times \frac{Y}{\Delta Y}$$

$$= \frac{30 - 25}{25} \times \frac{100}{102 - 100}$$

$$= \frac{5}{25} \times \frac{100}{2} = 10$$

$$\therefore E_Y = 10$$

Cross Elasticity of Demand is

Definition:

Cross elasticity of demand measures the responsiveness of the quantity demanded of one good (Good A) to a change in the price of another good (Good B).

Formula:

$$E_c = \frac{\text{change in quantity demanded of Good A}}{\text{change in price of Good B}}$$

Types of Cross Elasticity of Demand:

1) Positive Cross Elasticity ($E_c > 0$) → Substitute Goods

- If $E_c > 0$, the goods are substitutes.
- When the price of Good B rises, demand for Good A rises.

Example:

- Tea and coffee → If price rises, demand for tea increases.

2) Negative Cross Elasticity ($E_c < 0$) → Complementary Goods

- If $E_c < 0$, the goods are complements.
- When the price of Good B rises, demand for Good A falls.

Example:

- Cars and petrol → If petrol price rises, demand for car may decrease.

3) zero cross elasticity ($E_c = 0$) → unrelated goods

- If $E_c = 0$, the goods are unrelated.
- price change of good 'B' does not affect demand for good 'A'.

Example:-

- Salt and shoes → price of shoes does not affect demand for salt.

Example:

Calculate the cross elasticity b/w Samosa and Pakoda using following table.

Commodity	Before		After	
	Price	Quantity	Price	Quantity
Samosa x	20	1	20	2
Pakoda y	10	2	15	1

Solution:-

Initial demand for Samosa $Q_x = 1$

Initial price of Pakoda $P_y = 10$

change in demand for Samosa $\Delta Q_x = Q_{x1} - Q_x = 2 - 1 = 1$

change in price of Pakoda $\Delta P_y = P_{y1} - P_y = 15 - 10 = 5$

New demand for Samosa $Q_{x1} = 2$

New price of Pakoda $P_{y1} = \text{Rs. } 15$

We have

$$E_c = \frac{\Delta Q_x}{\Delta P_y} \times \frac{P_y}{Q_x} = \frac{1}{5} \times \frac{10}{1} = 2$$

$$\therefore E_c = 2$$

Principles of Engineering Economy

⇒ Engineering economy helps in choosing the best alternative among different technical or financial options.

1. > Develop the alternatives: ✓

↳ Engineering economy always begins with choices ✓
↳ (A decision is meaningful only when there are two or more alternatives.) ✓

Explanation:

- Every engineering problem can be solved in more than one way (e.g. different machines, designs, materials, or processes).
- The engineer must "identify, list, and develop all possible alternatives" before evaluating them.
- "Without alternatives, economic comparison cannot be done."

Example: choosing between a diesel generator and a solar power system for a factory.

2. > Focus on the differences: ✓

↳ Only the difference in cash flow, performance, cost and benefit between alternatives matter in economic comparison.

Explanation:

- Common or identical features of alternatives should be ignored. ✓
- Evaluation must be based on extra cost and extra benefit, also called incremental analysis. ✓
- This prevents unnecessary calculation and focuses on the real decision factors.

Example:

If both machines require the same installation cost, ignore it and compare only operating costs.

8) Use a Consistent viewpoint:

All decisions should be evaluated from a single, uniform perspective (e.g., owner's, viewpoint, company viewpoint, or customer viewpoint).

Explanation:

- Cash inflows and outflows must be considered from the same viewpoint, not mixed.
- This ensures fair comparison of alternatives.
- Using multiple viewpoints creates confusion and incorrect results.

Example:

If evaluating from the company's viewpoint, including labor cost, electricity, maintenance etc., but exclude personal or government benefits not affecting the company.

4) Use a Common unit of measurement:

All costs, benefits and consequences must be expressed using one uniform measurement unit, usually "money".

Explanation:

- Alternatives may have different parameters (energy, labor, time, etc.)
- To compare them fairly, convert everything to a single monetary unit (e.g., labor, electricity, etc.)
- This avoids confusion and makes alternatives comparable.

Example:

Machine 'A' uses less electricity but more labor; Machine 'B' uses more electricity but less labor. Convert both into Rs./month for comparison.

5) Consider All Relevant criteria:

Include every important cost, benefit, constraint, and effect associated with each alternative.

Explanation:

- Relevant criteria include initial cost, operating cost, resale value, maintenance, risk, life span, environmental impact etc.)
- Ignoring any important factor leads to wrong decisions.
- Both quantitative (monetary) and qualitative (safety, reliability) factors must be considered.

Example:

While selecting a vehicle, consider price, depreciation, maintenance, resale value, and consider not only the purchase

⇒ Make uncertainty explicit.

Every engineering project has some uncertainty (future costs, demand, price changes). These must be clearly stated.

Explanation:

- Future predictions are never 100% certain.
- Engineers must include uncertainty using sensitivity analysis, probability, or best-case/worst-case estimates.
- Including uncertainty explicit helps in preparing backup plans.

Example:

Electricity price may increase in the next 10 years → include this uncertainty when evaluating electric-powered machines.

7.7. Revisi your decision ✓

After selecting a preferred alternative, engineers must re-eval and update the decision when new information arises.

Explanation:

- Conditions may change (market price, technology, regulations).
- Re-evaluating ensures the decision stays optimal.
- It allows correction of mistakes and adaptation to unexpected situations.

Example:

If a company buys a diesel generator but later finds cheaper solar panels available, it should reconsider the previous decision.

Techniques of time value of money:

1.7. Future Value / Future Worth (FV):

$$FV = PV(1+i)^n \quad \text{--- (1)}$$

Where, $FV = \text{Future Value}$

$i = \text{Interest rate}$

$n = \text{number of years}$

$PV = \text{Present Value}$

Note: If bank follow compound interest → interest on interest.

Example: Mr. X deposits Rupees Rs. 50,000 now in

a bank account. How much money will be received at the end of 5th year if interest rate is 10%.

Solution:

Given,

$PV = \text{Rs. } 50,000$

$i = 10\% = 0.1$

$n = 5 \text{ year}$

$FV = ?$

We have,

$$FV = 50,000(1+0.1)^5$$

$$\therefore FV = \text{Rs. } 80,525$$

Alternative:

$$FV = PV \times FVIF_{10\%, 5 \text{ yr}}$$

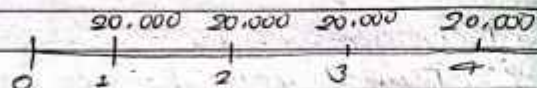
$FVIF = \text{Future Value Interest Factor}$

$$FV = 50,000 \times 1.6105$$

$$FV = \text{Rs. } 80,525$$

annuity -

- (i) Every year should deposit ✓
- (ii) deposit same amount ✓
- (iii) At the end of the year (deposit)



Formula is

$$(i) \quad FV = A \left[\frac{(1+i)^n - 1}{i} \right]$$

By using table

$$FV = A \times FVIFA_{i, n}$$

Example: Mr. A deposits Rs. 10,000 every year in a bank account. how much money will he get at the end of 5th year if interest rate is 10%.

Solution:

$$\text{Annuity (A)} = \text{Rs. } 10,000$$

$$\text{year} = 5 \text{ years (n)}$$

$$i = 10\% = 0.1$$

$$FV = 10,000 \left[\frac{(1+0.1)^5 - 1}{0.1} \right] = 61051$$

alternatively using financial table

(ii) present value of annuity (PV)

$$PV = A \left[\frac{(1+i)^{-n} - 1}{i(1+i)^n} \right]$$

$$PV = A \times PVIFA_{i, n}$$

Example: Mr. A needs Rs. 25,000 every year for 6 years. how much money does he have to deposit now to make this requirement if interest rate is 12%.

Solution:

$$A = \text{Rs. } 25,000$$

$$n = 6 \text{ year}$$

$$i = 12\% = 0.12$$

using formula

$$PV = 25,000 \left[\frac{(1+0.12)^{-6} - 1}{0.12(1+0.12)^6} \right]$$

$$PV = \text{Rs. } 102,785$$

using financial table

$$PV = 25,000 \times 4.1114$$

$$PV = \text{Rs. } 102,785$$

(3) present value / present worth (PV/PW):-

$$PV = \frac{FV}{(1+i)^n}$$

$$PV = FV(1+i)^{-n}$$

alternatively by using table

$$PV = FV \times PVIF_{i, n}$$

Example: Mr. Hart needs Rs. 5,00,000 at the end of 10 year for his daughter's education. How much money does he deposit now if interest rate is 12%?

Solution:-

$$FV = Rs. 5,00,000$$

$$PVIF = 3.1058$$

$$PV = 5,00,000 \times 3.1058$$

$$PV = FV (1+i)^{-n}$$

$$= 5,00,000 \cdot (1+0.12)^{-10}$$

$$PV = Rs. 1,60,986$$

Example: How much will I have to deposit for my university if interest rate is 10%?

Soln:-

Let suppose,

$$PV = Rs. 10,000$$

$$FV = Rs. 20,000 \text{ (double)}$$

$$i = 10\%$$

$$n = ?$$

We know,

$$FV = PV (1+i)^n$$

$$\Rightarrow 20,000 = 10,000 (1+0.1)^n$$

$$\Rightarrow \frac{20,000}{10,000} = (1.1)^n$$

$$\Rightarrow 2 = (1.1)^n$$

$$\Rightarrow n = 1.5176$$

$\therefore n \approx 2$ years

Calculation of PV for uneven cash flows:-

$$PV = \sum \left[\frac{C_t (1+i)^{-t}}{(1+i)^t} \right]$$

Example:-

$$FV = Rs. 10,000$$

$$i = 10\%$$

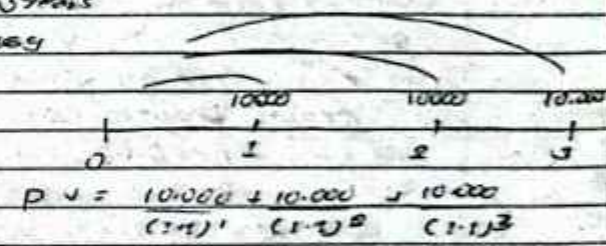
$$n = 3 \text{ years}$$

$$PV = ?$$

$$PV = \text{At PV of 10 years}$$

$$= 10,000 \times 2.4869$$

$$= Rs. 24,869$$



$$= Rs. 24,869$$

Example:- Calculating PV from the following cash flow

Year	1	2	3	4	5	6
CF	5000	8000	3000	2000	1000	6000

$$i = 10\%$$

$$PV = \frac{5000}{(1.1)^1} + \frac{8000}{(1.1)^2} + \frac{3000}{(1.1)^3} + \frac{2000}{(1.1)^4} + \frac{1000}{(1.1)^5}$$

$$+ \frac{6000}{(1.1)^6}$$

$$= Rs. 35,764$$

$$FV = 50,000 \times (1.1)^5 + 8000 \times (1.1)^4 + 3000 \times (1.1)^3 + 2000 \times (1.1)^2 + 1000 \times (1.1)^1 + 6000$$

$$Rs. 53,279$$

Ex: Bimla Gurung was student at the university in 2007, she borrowed Rs. 12000 at student loan at an annual interest rate of 9%. If she repays Rs. 1488 per year, how many 10-140 nearest year, will she take her loan repay the loan.

Given,

Loan Amount / Present Value = Rs. 12000
 Annual Payment (A) = Rs. 1488
 $i = 0.09$
 $n = ?$

We have,

$$PV = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

$$\Rightarrow (1+i)^n - 1 = \frac{PV \times i(1+i)^n}{A}$$

$$\Rightarrow (1+i)^n = \frac{PV \times i(1+i)^n}{A} + 1$$

$$\Rightarrow 1.09^n = \frac{12000 \times 0.09 \times (1.09)^n}{1488} + 1$$

$$\Rightarrow 1.09^n - 1 = 0.7258 \times (1.09)^n$$

$$n \log(1.09) = \log(0.7258) + n \log(1.09) = 1.988$$

$$\log(1.09)$$

$$\frac{\log(0.7258) - \log(1.09)}{0.09}$$

$$n = 35.0171$$

$$Ex: FV = ?$$

$$n = 3 \text{ years}$$

$$i = 10\%$$

$$A = 10,000$$

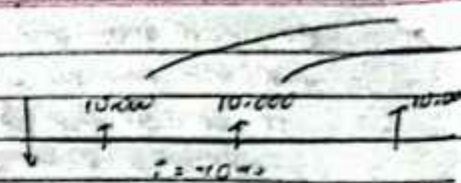
$$FV = A \times FVF_{i,n} = 3 \text{ years}$$

$$= 10,000 \times 3.3100$$

$$FV = Rs. 33,100$$

$$= 10,000 \times (1.1)^3 + 10,000 \times (1.1)^2 + 10,000 \times (1.1)^1$$

$$= Rs. 33,100$$



ii) Annuity due

c) Depositing amount at the beginning of the year

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right] (1+i)$$

Ex: Mr. Z needs Rs. 5,00,000 at the end of 10th year.

How much money need he to deposit every year if interest rate is 9%.

Solution -

$$FV = Rs. 5,00,000$$

$$\text{Year}(n) = 10$$

$$i = 9\% = 0.09$$

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right] (1+i)$$

$$A = \frac{FV \times i}{(1+i)^n - 1}$$

$$= \frac{5,00,000 \times 0.09}{(1.09)^{10} - 1}$$

$$A = Rs. 32,910$$

$$FV = 50,000 \times (1.1)^5 + 8000 \times (1.1)^4 + 3000 \times (1.1)^3 + 2000 \times (1.1)^2 + 1000 \times (1.1)^1 + 6000$$

$$Rs. 33,278$$

Ex: Amina Ghorng was student at the University in 2007, she borrowed Rs. 12000 at student loan at an annual interest rate of 9%. If she repays Rs. 1488 per year, how much will she have to pay at the end of 10 years?

Given,
Loan Amount (P) = Rs. 12000
Annual Payment (A) = Rs. 1488
 $i = 0.09$
 $n = 10$

We have,

$$PV = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

$$\Rightarrow (1+i)^n - 1 = \frac{PV \times i(1+i)^n}{A}$$

$$\Rightarrow (1+i)^n = \frac{PV \times i(1+i)^n}{A} + 1$$

$$\Rightarrow 1.09^n = \frac{12000 \times 0.09 \times (1.09)^n}{1488} + 1$$

$$\Rightarrow 1.09^n - 1 = 0.7258 \times (1.09)^n$$

$$n \log(1.09) = \log(0.7258) + n \log(1.09) = 1.925$$

$$PV(1+i)^n = A(1+i)^n$$

$$-A$$

$$PV(1+i)^n - A(1+i)^n = -A$$

$$\Rightarrow (1+i)^n = \frac{-A}{PV - A}$$

$$PV - A$$

$$\Rightarrow (1+i)^n = \frac{A}{A - PV}$$

$$A - PV$$

$$n = \frac{A}{A - PV}$$

$$\log(1+i)(A - PV)$$

$$\log(1.09)$$

$$\frac{1488 - 12000}{0.09}$$

$$\therefore n = 15.017$$

$$Ex: FV = ?$$

$$n = 3 \text{ years}$$

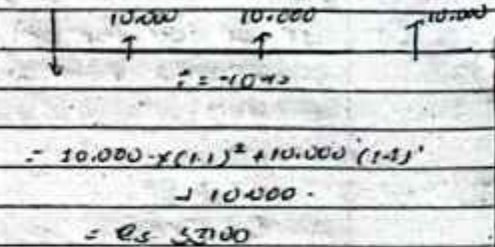
$$i = 10\%$$

$$A = 10,000$$

$$FV = A \times FVF_{i,n} = 3 \text{ years}$$

$$= 10,000 \times 3.3100$$

$$FV = Rs. 33,100$$



$$= 10,000 \times (1.1)^3 + 10,000 \times (1.1)^2 + 10,000 \times (1.1)^1 + 10,000$$

$$= Rs. 33,100$$

ii) Annuity due

Depositing amount at the beginning of the year

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right] (1+i)$$

Ex: Mr. Z needs Rs. 5,00,000 at the end of 10th year.

How much money now he to deposit at 10% interest rate is 9%.

Solution -

$$FV = Rs. 5,00,000$$

$$\text{Years } (n) = 10$$

$$i = 9\% = 0.09$$

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right] (1+i)$$

$$A = \frac{FV \times i}{(1+i)^n - 1}$$

$$(1+i)^n - 1$$

$$= \frac{5,00,000 \times 0.09}{(1.09)^{10} - 1}$$

$$A = Rs. 32,910$$

Ex: How much money will Shyam receive at the end of 8 year if he deposits 10,000 at the beginning of each year if interest rate is 8%.

Solution:

Given

$$r = 0.08$$

$$n = 8 \text{ yrs}$$

$$P = \text{Rs. } 20,000$$

Since, the amount is deposited at the beginning of every year i.e. Rs. 10,000 at the rate of 8% p.a.

We know,

$$FV = P \left[\frac{(1+r)^n - 1}{r} \right] (1+r)$$

$$\text{Rs. } 31,987.5$$

Types of Interest rate

1) Simple Interest rate

$$\text{Ex: } PV = \text{Rs. } 10,000$$

$$r = 10\%$$

$$n = 5 \text{ yrs}$$

$$FV = ?$$

We know

$$FV = 10,000 \times 0.1 \times 5 + 10,000$$

$$= 50,000 + 10,000$$

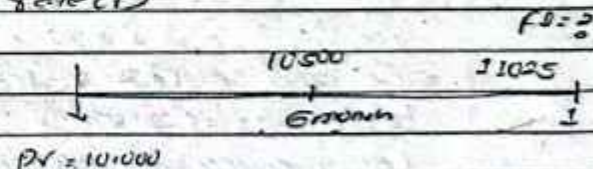
$$= \text{Rs. } 150,000$$

(ii) Compound Interest rate (one time interest - annual compound)

$$FV = PV (1+r)^n$$

(iii) Effective Interest rate (EIR)

Ex:-



$$\text{Effective Interest rate} = \frac{11,025 \times 100}{10,000}$$

$$= 110.25\%$$

$$\text{Effective Interest rate (EIR)} = (1 + \frac{r}{m})^m - 1$$

where, m = number of compounding period in a year

$$r = (1 + 0.1)^2 - 1$$

$$r = 10.25\%$$

Ex. Mr. Hari deposits Rs. 200000 in a bank. How much money he gets at the end of 8th year (1997-2000) interest rate:

(a) Following simple interest

(b) Following annually compounding

(c) Following quarterly compounding

(d) Following simple interest rate

$$FV = P + N \cdot P \cdot R$$

$$= 200000 + 8 \times 200000 \times 0.12$$

$$= \text{Rs. } 392000$$

(e) Following annually compounding

$$FV = 200000 (1 + 0.12)^8$$

$$FV = \text{Rs. } 519510$$

Since the interest is calculated on the basis of quarterly compounding
Let's calculate effective interest rate

$$(1 + \frac{0.12}{4})^4 - 1$$

$$0.1255$$

$$12.55\%$$

Now,

calculation for FV

$$FV = P (1 + R)^N$$

$$= 200000 (1 + 0.1255)^8$$

$$= \text{Rs. } 519580$$

Alternatively,

$$FV = P (1 + R)^N$$

$$FV = P (1 + \frac{R}{m})^{N \cdot m}$$

$$= 200000 (1 + \frac{0.12}{4})^{32}$$

$$= \text{Rs. } 515010$$

EMI:

$$\text{Loan amount} = \frac{E \times (1+i)^n - 1}{i(1+i)^n}$$

$$\text{Loan amount} = \frac{E \times (1+i/m)^{n \times m} - 1}{i/m \times (1+i/m)^{n \times m}}$$

Example: - Mr. X is willing to buy a new car costing Rs. 50,00,000. He applied for a loan at NIMB Ltd. The bank asked him to arrange some cash as down payment. How much money does Mr. Pandey need to pay every month for 5 years to settle this loan if interest rate is 8%.

Solution:-

$$\text{Total amount} = \text{Rs. } 50,00,000$$

$$\text{Cash down payment percentage} = 20\%$$

$$\text{So, Loan amount} = 50,00,000 - \frac{50,00,000 \times 20}{100}$$

$$= \text{Rs. } 50,00,000 - 10,00,000$$

$$= \text{Rs. } 40,00,000$$

$$EMI = ?$$

$$n = 5$$

$$m = 12$$

$$i = 8\%$$

Now, we will

$$EMI = \frac{\text{Loan amount} \times i/m \times (1+i/m)^{n \times m}}{(1+i/m)^{n \times m} - 1}$$

$$= \text{Rs. } 81105$$

Alternative:

By using monthly effective interest rate.
We have,

$$\begin{aligned} i_{\text{monthly}} &= (1+i)^{1/12} - 1 \\ &= (1+0.08)^{1/12} - 1 \\ &= 0.00693 \end{aligned}$$

$$\text{Known amount} = EMI \times \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

$$EMI = \frac{40,00,000 \times 0.00693 \times (1+0.00693)^5}{(1+0.00693)^5 - 1}$$

$$= \text{Rs. } 8159.97$$

PV - Asked is interest

How many deposit of Rupees Rs. 25000 each should Mr. Shukla make each month so that the final accumulated amount will be 150,000, if bank interest is 12%.

Solution:-

$$FV = \text{Rs. } 150,000$$

$$i = 12\% = 0.12$$

$$\text{Monthly deposit} = \text{Rs. } 25000$$

$$n = ?$$

We know,

$$FV = P \left[\frac{(1+i)^n - 1}{i/m} \right]$$

$$\Rightarrow \frac{10,00,000}{25000} = \left[\frac{(1+0.12/12)^n - 1}{0.01} \right]$$

$$\Rightarrow 0.4 + 1 = (1+0.01)^n$$

$$\Rightarrow 1.4 = (1.01)^n$$

$$n = \frac{10 \times (1.12)}{(1.01)}$$

$$\therefore n = 30 \text{ times}$$

Answer 1

$$\begin{aligned} \ln &= (1+i)^{1/12} - 1 \\ &= (1.00978)^{1/12} - 1 \\ &= 0.00978 \end{aligned}$$

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right]$$

$$10,00,000 = 25000 \left[\frac{(1.00978)^n - 1}{0.00978} \right]$$

$$\Rightarrow \frac{10,00,000}{25000} \times 0.00978 = (1.00978)^n - 1$$

$$\Rightarrow (1.00978)^n = 1.3792$$

$$\Rightarrow n = \frac{\log(1.3792)}{\log(1.00978)}$$

$$n = 34$$

Question

Same amount of same percentage.

1.2 Arithmetic Gradient (Uniform Gradient)

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right] + G \left[\frac{(1+i)^n - 1}{i^2} \right] - ng$$

A = First year's deposit

G = Incremental amount (every year)

$$PV = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right] + G \left[\frac{(1+i)^n - 1 - n}{i^2(1+i)^n} \right]$$

Problem:

A person P is planning for retired life and has 10 more years to retire. i.e. would like to deposit some portion of his salary, which is Rs. 5000 at the end of first year and then after he wishes to increase the amount with Annual deposit of Rs. 1000 for next 9 years with an interest of 15% p.a. Total amount at the end of 10th year.

Solution:

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right] + G \left[\frac{(1+i)^n - 1}{i^2} \right] - ng$$

$$= 5000 \left[\frac{(1+0.15)^{10} - 1}{0.15} \right] + 1000 \left[\frac{(1+0.15)^{10} - 1}{0.15} \right] - 10 \times 1000$$

$$= \text{Rs. } 108387.$$

ANNUAL

TRICK:-

Year	1	2	3	4	5	6	7	8	9	10
C.F	5000	6000	7000	8000	7000	8000	9000	10000	11000	12000

$$FV = 5000 \times (1.15)^9 + 6000 \times (1.15)^8 + 7000 \times (1.15)^7 + 8000 \times (1.15)^6 + 7000 \times (1.15)^5 + 8000 \times (1.15)^4 + 9000 \times (1.15)^3 + 10000 \times (1.15)^2 + 11000 \times (1.15)^1 + 12000$$

(2.) Geometric Gradient:

• For $r \neq g$:

$$PV = \frac{A}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^n \right]$$

where, A = first cash flow.

r = interest rate.

g = growth rate.

n = no. of year.

• For $r = g$:

$$PV = \frac{A \times n}{1+r}$$

Ex: A Company expect to receive a series of payments starting at ₹5,000 per year one, growing at 4% per year for 10 years if the discount rate is 7%, what is the present value of these cash flow.

Given:

$$g = 4\% = 0.04$$

$$r = 7\% = 0.07$$

$$A = ₹5,000$$

$$PV = ?$$

we have,

$$PV = \frac{5000}{0.07 - 0.04} \left[1 - \left(\frac{1 + 0.04}{1 + 0.07} \right)^{10} \right]$$

₹5,91253.

• FUTURE VALUE:-

• YF $r \neq g$

$$FV = AX \left[\frac{(1+r)^n - (1+g)^n}{r-g} \right]$$

• YF $r = g$

$$FV = A \times n \times (1+r)^{n-1}$$

Q. 100 ASKED:-

Ex: Ramash, a computer engineer is planning to deposit
 90% of his total salary which is
 Rs. 250,000 per year, each year in mutual
 funds. He expects 7% salary increase for
 each year for 15 years. If the mutual funds will
 give 10% annual return what will the sum
 amount at the end of 15 years? And if salary
 increases by Rs. 25,000 per year what will be
 the amount

Solution:-

1st case:-

Salary increasing 7% every year

assuming depositing at the end of every year.

1. $250000 \times 1.07 = \text{₹ } 267500$ end year salary Deposit 2011 → 250000

2. $267500 \times 1.07 = 286225$ Deposit 2012 → 267500

8

3. $286225 \times 1.07 = 306260$ Deposit 2013 → 286225

% difference = $\frac{57295 - 53500}{53500} \times 100 = 7\% \checkmark$
 53500

$A = 53500$ ✓

$g = 7\%$ ✓

$r = 10\%$

$FV = ?$, $n = 15 \text{ yrs}$

now,

~~for~~

$$FV = 53500 \times \left[\frac{(1+0.1)^{15} - (1+0.07)^{15}}{0.1 - 0.07} \right]$$

$= \text{Rs. } 6848976$

$\text{Rs. } 25,29,152$

2nd case:

Salary increasing by Rs. 25000 every year.

assuming depositing at the end of every year.

1st year salary $(25000 + 25000) \rightarrow 50000$ ✓ 5000

2nd year salary $(50000 + 25000) \rightarrow 75000$ ✓ 60000

3rd year salary $(75000 + 25000) \rightarrow 1,00,000$ ✓ 65000

Since, the amount of deposit is increased by

5000 every year. It is a case of Arithmetic

gradient.

where

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right] + G \left[\frac{(1+i)^n - 1}{i} \right] - ng$$

$$= 55000 \left[\frac{(1+0.1)^{15} - 1}{0.1} \right] + 5000 \left[\frac{(1+0.1)^{15} - 1}{0.1} \right] - 15 \times 5000$$

$\text{Rs. } 25,86,110$

What happens if he deposits now (beginning of the year)

first case:

first year 250000 Deposit 2010 → 50,000
 second year $(250000 \times 1.07) = 267500 \rightarrow$ Deposit 2011 → 53500
 3rd year $(267500 \times 1.07) = 286225 \rightarrow$ Deposit 2012 → 57245

Ex: Mr. Adhikary wants to have Rs. 30,00,000 for the
 studies of his son after a period of 15 years.
 How much approx does he have to deposit end
 of each year for 10 continuous year in saving
 account that earns 10% yearly annuity.

Solution:-

FV = Rs 30,00,000

no. of year = 15 year

FV = 30,00,000

Interest rate = 10%

= 0.1

10 5 year 15

every year deposit for 10 year?

Step 1:-

Discounting 15 years value i.e. 30,00,000 into 10th year.

Value of 10 year = $\frac{30,00,000}{(1.1)^5}$ = Rs. 18,62,763

Step 2:- Calculation of every year deposit for 10 year
by using the following formula.

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right]$$

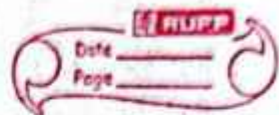
$$= 18,62,763 = A \left[\frac{(1.1)^{10} - 1}{0.1} \right]$$

$$\Rightarrow A = \frac{18,62,763 \times 0.1}{(1.1)^{10} - 1}$$

$$A = \text{Rs. } 11,6570$$

$$\text{Rs. } 11,6570$$

Depreciation



Fixed Assets

CAUSES OF DEPRECIATION → Important (5 marks)

Terminology:

① Purchase price → zero year, original cost - the cost of the assets

② Annual depreciation amount

③ Book value → value of assets after deducting annual

depreciation
first year → purchase price - annual depreciation.

4) Salvage value → value of an assets in nth year.

5) no. of years (n)

6) Depreciation amount methods

Types:

1) Straight line depreciation method:

$$\text{Annual depreciation (AD)} = \frac{\text{Cost of assets} - \text{Salvage value}}{\text{no. of years (useful life)}}$$

Ex: → A machine costing of Rupees Rs. 120,000 having estimated salvage value of Rupees Rs. 20,000 and 10 years useful life. Calculate annual depreciation amount and book value on straight line depreciation method.

Solution:

Calculation of Annual depreciation Amount (AD)

We have

$$\text{AD} = \frac{\text{Cost of machine} - \text{Salvage value}}{\text{no. useful life}} = \frac{120,000 - 20,000}{10} = 10,000$$

Calculation of Annual Book Value

Year	Beginning book value	Annual depreciation	Ending Book value
1	120,000	10,000	110,000
2	110,000	10,000	100,000
3	100,000	10,000	90,000
4	90,000	10,000	80,000
5	80,000	10,000	70,000
6	70,000	10,000	60,000
7	60,000	10,000	50,000
8	50,000	10,000	40,000
9	40,000	10,000	30,000
10	30,000	10,000	20,000

2. Diminishing value method / declining value method :-

Ex:- A machine costing of Rs. 20,000 having estimated
 salvage value of Rupees Rs. 3,000 and
 6 year useful life. Calculate annual
 depreciation value and book value. Under
 Diminishing value ~~method~~ balance with 20%
 depreciation rate.

	Beginning	Annual	End
Year	Book Value	Depreciation	Book Value
1	200000	40000	160000
2	160000	32000	128000
3	128000	25600	102400
6	102400	20480	81920

✓ Diminishing balance method (where rate is not given) :-

$$\text{Rate (R)} = \left[1 - \sqrt[n]{\frac{S}{\text{original cost}}} \right] \times 100$$

Ex: A machine cost Rs. 36,000 and estimated salvage value at the end of 5th year is supposed to be 6,000 calculate annual depreciation amount and book value under diminishing balance method.

(Since,

rate is not given,
we have

$$R = \left[1 - \sqrt[5]{\frac{6000}{36000}} \right] \times 100$$

$$= 30.12\%$$

Calculation of depreciation amount and book value,

year	beginning book value	Depreciation at rate of 30.12%	ending book value
1	36000	10893.2	25156.8
2	25156.8	7577.23	17579.57
3	17579.57	5299.96	12289.60
4	12289.60	3700.12	8589.47
5	8589.47	2585	5998.82

3) Sinking Fund Method:

Step 1: Calculation of fixed depreciation amount with the help of following formula.

$$\text{Fixed depreciation} = \frac{I - S}{\text{Life in years}}$$

Ex: Compute the depreciation charge and book value of each year by using sinking fund method with following info.

Initial cost of the Asset = Rs. 100000 (I)

Salvage value = 20,000 (S)

Life of Asset = 8 year (n)

Interest rate = 12% (i)

Solution:

Step 1: Calculation of fixed depreciation amount.

$$\begin{aligned}
 \text{Fixed depreciation} &= \frac{I - S}{\text{Life in years}} \\
 &= \frac{100000 - 20000}{12.2897} \\
 &= 6509.22
 \end{aligned}$$

Step 2: Calculation of every year depreciation amount and book value

Year	Beginning book value	Fixed depreciation	Net depreciation = $FD \times (1+i)^{n-1}$	Ending book value
1	100000	6509.22	6509.22	93495.78
2	93495.78	6509.22	7284.72	86211.06
3	86211.06	6509.22	8158.89	78052.17
4	78052.17	6509.22	109657.67	

- | | |
|----|----------|
| 5. | 9137.06 |
| 6. | 10239.51 |
| 7. | 11962.65 |
| 8. | 12868.17 |

in there

(K = particular year)

$$d_k = FD \times FV \text{ if } i\% (k-1) \text{ year}$$

where,

$$FD = \left[\frac{I - E}{FV \text{ if } i\% n \text{ years}} \right] \times FV \text{ if } i\% (k-1) \text{ year}$$

Depreciation of 7th year :-

$$\begin{aligned} d_7 &= 6509.22 \times FV \text{ if } 12\% 6 \text{ year} \\ &= 6509.22 \times 1.0738 \\ &= 12808.02 \end{aligned}$$

Calculation ending book value for the particular year

$$BV_k = I - [FD \times FV \text{ if } i\% k \text{ years}]$$

$$\begin{aligned} BV_k &= 100000 - [6509.22 \times 2.0710.0890] \\ &= 89878.92 \end{aligned}$$

4. Sum of year digits method (SOYD) :-

Ex: A machine costing of Rs. 1,50,000 having estimated salvage value of Rs. 15,000 and 10 years useful life. Calculate annual depreciation amount and book value after SOYD method.

Solution: $I = 1,50,000$

Given, $S = 15,000$ SOYD = 2

Step 1: Calculation of SOYD :-

$$\text{SOYD} = \frac{n(n+1)}{2} = \frac{10(10+1)}{2} = 55$$

Step 2: Calculation of Annual depreciation and book value
 $= (I - S) \times \frac{\text{year}}{\text{SOYD}}$

Year	Beginning book value	Annual depreciation	End of book value
1	150000	13181.81	13181.81 138181.20
2	150000	10636.36	133636.37
3		8454.54	135454.55
4		6909.09	139090.92
5		5909.09	140909.2
6		4727.27	142727.28
7		3545.45	144545.46
8		2536.36	146363.64
9		1818.18	148181.82
10			

5> double rate ^{declining} depreciation method & double rate declining method:

Step 1:- calculation of depreciation rate (rate of straight line).

$$\text{Rate} = \left(\frac{1}{n} \times 100 \right) \times 2$$

If n is 10 years

$$r = \frac{1}{10} \times 100 \times 2 = 20\%$$

Step 2: Use diminishing balance method with above rate and calculate depreciation amount and book value.

Ex: Consider the following account info for computer system.

Initial cost of the computer = Rs. 10,000

Useful life = 5 years

Compute the annual depreciation amount and remaining book value by using double rate declining method.

Step 1: calculation of rate:

$$r = \left(\frac{1}{n} \times 100 \right) \times 2 = \frac{1}{5} \times 100 \times 2 = 40\%$$

Year	Beginning Book Value	Depreciation at rate 40%	Ending Book Value
1	10,000	4,000	6,000
2	6,000	2,400	3,600
3	3,600	1,440	2,160
4	2,160	864	1,296
5	1,296	518.4	777.6

67. Straight line + interest method (interest method):

Ex: Consider the following machine operating system.
 Initial cost = Rs 4000
 Useful life = 10 yrs
 Salvage value = 0
 depreciation rate
 Compute the annual depreciation book value by using interest method.

Calculation of depreciation

Year	Beginning Book Value	Dep. = $\frac{\text{Book Value}}{\text{SL Period}}$	Dep. DB
1	4000	400	3600
2	3200	320	2880
3	2560	256	2304
4	2048	204.8	1843.2
5	1638.4	163.84	1474.56
6	1310.72	131.072	1179.648
7	1048.576	104.8576	943.7184
8	833.664	83.3664	750.2976
9	650.2976	65.02976	585.26784
10	500.26784	50.026784	450.241056

67 Straight line + average interest method / SLD method
method (important):

Ex: Consider the following information for a machine operating system

Initial cost = Rs 4000

Useful life = 10 yrs

(Salvage value) = 0

Depreciation rate = 20%

Compute the annual depreciation amount and book value by using straight line + average interest method.

Calculation of depreciation amount and book value

Year	Beginning book value	Dep. = $\frac{\text{Remaining balance}}{\text{SL method}} \times \text{rate}$	Dep. DB method @ 20%	Dep. Separation	Ending book value
1	4000	400	800	800	3200
2	3200	355.55	640	640	2560
3	2560	320	512	512	2048
4	2048	292.57	409.6	409.6	1638.4
5	1638.4	273.06	327.6	327.6	1310.8
6	1310.8	262.16	262.16	262.16	1048.64
7	1048.64	262.16	209.72	262.16	786.48
8	786.48	262.16	157.292	262.16	524.32
9	524.32	262.16	104.864	262.16	262.16
10	262.16	262.16	52.432	262.16	0

7. Service Output Method / Unit of Production Method

Ex: → A machine costing of Rs. 50,000 is expected to have Rs. 10,000 salvage value when replaced after 30,000 hours of use. Find its depreciation rate per hour, its use and depreciation amount and book value after 10,000 hours of operation.

(Solution):

Calculation of rate.

$$\begin{aligned}
 &= \frac{\text{Cost} - \text{Salvage}}{\text{Life}} = \frac{50,000 - 10,000}{30,000} \\
 &= 1.33
 \end{aligned}$$

Now, depreciation amount after use of 10,000 hours

$$10,000 \times 1.33$$

$$= 13,300$$

$$\begin{aligned}
 \text{Book Value} &= \text{Cost Price} - \text{Depreciation} \\
 &= 50,000 -
 \end{aligned}$$

Cost: Rs. 50,000

SV = 10,000

Total hours = 30,000

Used hours = 10,000

Dep rate = ?

Dep amount = ?

Book value

$$\begin{aligned}
 \text{Rate} &= \frac{\text{Cost} - \text{SV}}{\text{No. of hours}} = \frac{50,000 - 10,000}{30,000} \\
 &= 1.33
 \end{aligned}$$

Dep. amount after 10,000 hours

$$= 10,000 \times 1.33 = 13,300 //$$

Book Value = Cost - Dep

$$= 50,000 - 13,300$$

$$= 36,700$$

Depreciation :-

Depreciation is the gradual and permanent decrease in the value of a fixed asset due to wear and tear, aging, or obsolescence.

Causes of depreciation :-

1. Wear and tear :-

→ continuous use reduces the asset's efficiency and value.

2. Passage of time :-

→ Some assets lose value simply due to aging, even without use.

3. Obsolescence :-

→ New technology or style make existing assets outdated.

4. Exhaustion / Depletion :- Natural resource like mines reduce in quantity when used.

5. Accidents :-

→ Damage from natural or man-made accidents lower the assets value.

6. Fall in market value :-

→ asset value drops when market prices decline.

Chapter 1
 Depreciation

EMI (Equal Monthly Installment) :-
 EMI is a fixed monthly payment made by a
 borrower to repay a loan, which includes
 both principal and interest, paid over a
 specific period of time.

8> The modified accelerated cost recovery system
 (MACRS) :-

Depreciation → 100%	3	} SL: $\frac{1}{n} \times 100$, DB: 2
1 yr 5 years	5	
10 yr	7	
10 yr → 0.5 (limit)	10	
1	} Full	Note: DB: $LSY (1.5)$
3		
4		
5		
6 yr → 0.5 (limit)		

(4-asset)

Ex: A tax payer wants to place in service of
 Rs. 10,000 Assets i.e. assigned to 140 5 year
 category 140 every year in MACRS percentage,
 depreciation amount and book value of assets.

Soln:
 Calculation of MACRS %

Rough:

$$SL \text{ rate} = \frac{1}{n} \times 100 = \frac{1}{5} \times 100 = 20\%$$

$$DB \text{ rate} = 20 \times 2 = 40\%$$

Year	Minerals Acquisition	Depreciation for Exploration
(1) 1	SL $\rightarrow \frac{1}{10} \times 100 \times 0.5 = 5.00$ DB $\rightarrow \frac{1}{10} \times 100 \times 2 \times 0.5 = 10.00$	20.00
2	SL $\rightarrow \frac{1}{10} \times 80 = 8.00$ DB $\rightarrow 80 \times 0.5 = 40.00$	32.00
3	SL $\rightarrow \frac{1}{10} \times 78 = 7.80$ DB $\rightarrow 78 \times 0.5 = 39.00$	40.20
4	SL $\rightarrow \frac{1}{10} \times 76 = 7.60$ DB $\rightarrow 76 \times 0.5 = 38.00$	47.80
5	SL $\rightarrow \frac{1}{10} \times 74 = 7.40$ DB $\rightarrow 74 \times 0.5 = 37.00$	55.20
(1) 6	SL $\rightarrow \frac{1}{10} \times 72 = 7.20$ DB $\rightarrow 72 \times 0.5 = 36.00$	62.40

Special part:

Calculation of Annual Depreciation and Book Value

Year	Original Cost	Minerals	Dep. Amount	Ending B.V.
1	10000	20.00	2000	8000
2	8000	32.00	3200	4800
3	7800	40.20	4020	3780
4	7600	47.80	4780	2820
5	7400	55.20	5520	1880
6	7200	62.40	6240	960

Ex: A ~~private~~ company bought ~~ore~~ land ~~in~~ ~~the~~ ~~year~~ ~~2000~~ for Rs. 20,00,000. The Reserve Reserves in the mine were estimated to be 5,00,000 tons.

(a) If 50,000 tons of ore were mined and sold during the year, what was the depletion charge and ending book value for 2000?

Solution:

Calculation of Cost per ton

$$\text{Per-ton cost} = \frac{\text{Total cost}}{\text{No. of tons}} = \frac{20,00,000}{5,00,000} = \text{Rs. 4 per ton}$$

Calculation of depletion amount for 50,000 tons.

$$= 50,000 \times 4 = \text{Rs. 2,00,000}$$

$$\text{Ending book value at the end of 2000} = 20,00,000 - 2,00,000 = \text{Rs. 18,00,000}$$

(b) Suppose at the end of 2000 the Reserves were re-valued, and found to be only 4,00,000 tons. If 50,000 were mined and sold in during 2001 what is depletion amount for 2001 and book value.

Soln:-

$$\begin{aligned} \text{Calculation of cost per ton} &= \frac{\text{Total cost}}{\text{No. of tons}} = \frac{18,00,000}{4,00,000} \times 50,000 \\ &= 4.5 \times 50,000 \\ &= 2,25,000 \end{aligned}$$

* Effective Interest Rate (EIR)

↳ The Effective Interest Rate (EIR) is the true annual rate of interest that shows how much interest is actually earned or paid in one year when compounding is considered. (It is more accurate than the nominal (stated) interest rate because it includes the effective compounding periods such as monthly, quarterly or daily.)

↳ When interest is compounded more frequently, the effective interest rate becomes higher than the nominal rate. Therefore, EIR helps in comparing different investments or loan options that have different compounding periods.

The formula for effective interest rate is:

$$i_{\text{effective}} = \left(1 + \frac{i}{m}\right)^m - 1$$

Where,

- i = nominal annual interest rate.
- m = number of compounding periods per year.

Q. Mr. Hari Lamsal needs Rs. 2500000 at the end of 15th year in his bank account. How much money does he need to deposit at the end of every year for ~~ten~~ 10 years to meet his requirement if his bank provides him 10% interest compounding annually?

Solution:

Step 1: Discounting the 15 years Rs. 2500000 into 10th year
Value of 10 year = $\frac{2500000}{(1.12)^5}$

$$= 1418567.135$$

Step 2: Now, calculation of every 6 year deposits for 10 years.

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right]$$

$$\Rightarrow 1418567 = A \times \left[\frac{(1.12)^{10} - 1}{0.12} \right]$$

$$\Rightarrow A = \frac{1418567 \times 0.12}{(1.12)^{10} - 1}$$

$$\therefore A = \text{Rs. } 80865.83$$

15) You make 10 annual deposits of 1,000 each into a bank account paying 5% interest per year. The first deposit will be made one year from today. How much money can be withdrawn from this account after 10 years? Initial 0 is the present worth of the sum.

Solution:-

Annual deposit = Rs. 1,000

$n = 10$ years

$i = 5\% = 0.05$

Now,

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right] \\ = 1,000 \left[\frac{(1.05)^{10} - 1}{0.05} \right] \\ = \text{Rs. } 12,577.89$$

16) You deposited Rs. 50,000 at the beginning of each for 7 years. How much money will be in your account at the end of 10th year when rate of interest is 6% compounded quarterly. Write and cash flow diagram.

Solution:-

$$\text{Effective interest rate} = (1 + i/m)^m - 1 \\ = (1 + 0.06/4)^4 - 1 \\ = 0.0613 = 6.13\%$$

The future value at the end of year m of an annuity-due with payments A for m periods is,

$$FV = A \left[\frac{(1+i)^m - 1}{i} \right] (1+i)$$

$$= 10,000 \left[\frac{(1+0.00613)^7 - 1}{0.00613} \right] (1+0.00613)$$

$$= 3,586.88$$

Since amount at the end of 7 years is the beginning of 10th + 0 to 10 or 13 more years.

$$\therefore FV_{10} = FV_7 \times (1+i)^3 \\ = 3,586.88 \times (1+0.00613)^3 \\ = 3,608.86$$

Q. El. Ram prasad deposited Rs. 1,00,000 in his bank account on 1 January 2022. The bank pays 9% simple interest. How much will he accumulate in his account on 30 Dec 2023?

155 If you deposit Rs. 5,000 per month for two years, what will be the amount at the end of five years if bank interest rate is 5% in every six months?

Given,

Monthly deposit, $A = 5000$

Deposit period = 2 years (24 months)

Total period = 5 years (60 months)

Interest rate = 5% p.a. 6 months

$FV_{60} = ?$

Step 1: Monthly interest rate i_m

$$i_m = (1+i)^{1/6} - 1$$

$$= (1.05)^{1/6} - 1 \approx 0.0081699$$

Step 2: Future value of deposits at month 24

$$FV_{24} = A \left[\frac{(1+i_m)^{24} - 1}{i_m} \right]$$

$$= 5000 \times \frac{1.215506 - 1}{0.0081699} \approx 5000 \times 26.399 \approx 131995$$

Step 3: Compound FV_{24} to month 60.

Remaining 36 months = 6 periods of 6 months at 5% each.

$$FV_{60} = FV_{24} \times (1.05)^6$$

$$= 131995 \times 1.276282 \approx 168425$$

Q. Suppose a farmer wants to save money semi-annually in a financial company for the engineering education of his daughter of 2 years old. How much money does he need to save per period if she will need 20,00,000 when her age will be 18 years old? The company compounds the money semi-annually and interest rate is 12%.

Solution:-

$$i = 0.12$$

$$FV = 20,00,000$$

$$\begin{aligned}
 n &= 18 \div 2 = 9 \times 2 \text{ (Semiannually)} \\
 &= 18 \text{ (m \times n)}
 \end{aligned}$$

Now we know,

$$FV = A \left[\frac{(1 + i/m)^{m \times n} - 1}{i/m} \right]$$

$$\Rightarrow 20,00,000 = A \left[\frac{(1.06)^{18} - 1}{0.06} \right]$$

$$\Rightarrow A = \frac{20,00,000 \times 0.06}{(1.06)^{18} - 1}$$

$$= 1859.4$$

Q(a) You have taken an auto loan of Rs. 2,20,000 from a bank, which is to be repaid in equal-end of month installments for 3 years with nominal interest rate of 15 percent compounded monthly. Calculate:

- i) the amount of each installment.
- ii) effective rate of interest loan.

Solution:

Loan Amount P : Rs. 220,000

Term: 3 years (n)

i : 15%

Time: 3

Now we know

$$\text{Loan Amount} = EMI \times \left[\frac{(1 + i/m)^{nm} - 1}{i/m (1 + i/m)^{nm}} \right]$$

$$\Rightarrow EMI = \frac{220,000 \times 0.15/12 (1 + 0.15/12)^{3 \times 12}}{(1 + 0.15/12)^{3 \times 12} - 1}$$

$$= \text{Rs. } 7626.$$

ii) effective annual interest rate

$$i_{\text{eff}} = (1 + i/m)^m - 1$$

$$= (1 + \frac{0.15}{12})^{12} - 1$$

$$0.1607$$

$$= 16.08\%$$

2a) Sabina deposits a sum of Rs. 10,00,000 in a bank at an interest rate of 12% per year. What will be the future amount after 5 years if compounded.

12 weekly

12 quarterly and annually.

Solution

12 weekly compounding

$$m = 52 \text{ weeks/year}$$

$$n = 5$$

$$FV = PV \times (1 + i/m)^{n \times m}$$

$$= 10,00,000 \times (1 + 0.12/52)^{5 \times 52}$$

$$= 1820855$$

2) Quarterly

$$m = 4$$

$$FV = PV \times (1 + i/m)^{n \times m}$$

$$= 10,00,000 \times (1 + 0.12/4)^{20}$$

$$= \text{Rs } 180611$$

3) Annually, $m = 1$

$$FV = PV (1 + i)^n$$

N.V.S

Basic methods of Engineering Economics study:-

1. > Equivalence worth method

- (a) Present worth (PW)
- (b) Future worth method (FW)
- (c) Annual worth method (AW)

2. > Rate of Return method — (1)

- (a) IRR method
- (b) ERR method

3. > Payback period method — (1)

- (a) simple ✓
- (b) Discounted ✓

4. > Benefit cost ratio method — (1)

- (a) conventional BC ratio ✓
- (b) modified BC ratio ✓

Parameters of project is

- 1) Initial investment (P) (one time at beginning)
- 2) Annual revenue (AR) (annuity) (every year)
- 3) Annual cost (AC) (annuity) every year.
- 4) Net Annual Revenue (NAR) (annuity) every year

$$\Rightarrow AR - AC$$

5. > Salvage value : $\rightarrow$ f.v. at the end of project life.

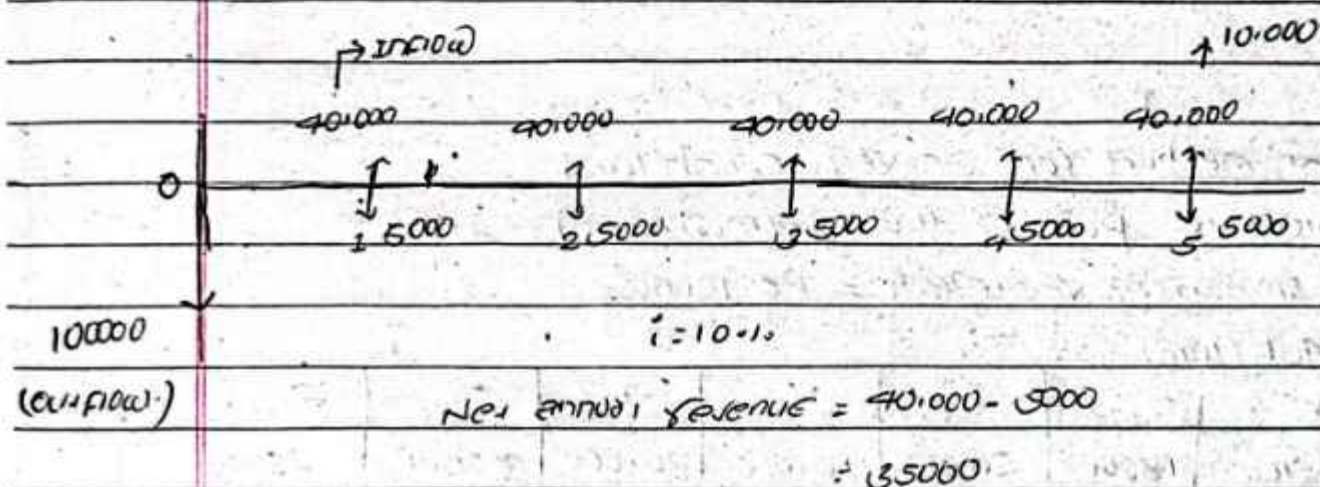
6. > Useful life : $\rightarrow$ n years.

7. > MARR $\rightarrow$ interest rate,

(minimum attractive rate of return)

Cash flow diagram:

- (a) Initial investment = 100000
- (b) Annual Revenue = 40000
- (c) Annual cost = 5000
- (d) Useful life = 5 years
- (e) Salvage value = 10000
- (f) $i = 10\%$



17. Equivalent worth methods-

a. > PW method:-

$$PW_{10\%} = -I + NAR \times PVIFA_{10\%, 5 \text{ yrs}} + SV$$

$$= -100000 + 35000 \times 3.7908 + \frac{10,000}{(1+0.1)^5}$$

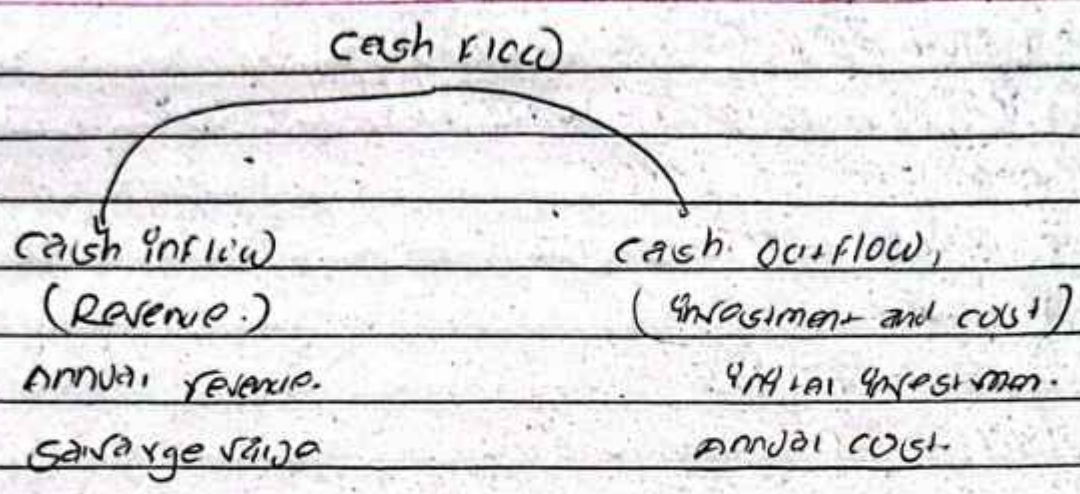
$$= +3887.21$$

Decision Rule under PW:-

- 1) Accept the project if PW is positive.
- 2) Reject the project if PW is -ve.

Since, the PW of the project is +ve, the project should be accepted.

Q.11

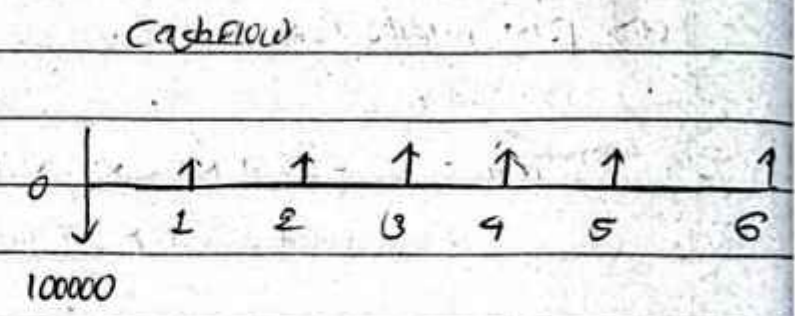


PW method for uneven cashflow:
 Calculate PW of the following project
 Initial investment = Rs. 100000

Cashflow:

Year	1	2	3	4	5	6
Cashflow	18000	20000	40000	30000	35000	15000

Interest rate (MARR) = 10%



$$\begin{aligned}
 PW_{10\%} &= -100000 + \frac{18000}{(1.1)^1} + \frac{20000}{(1.1)^2} + \frac{40000}{(1.1)^3} + \frac{30000}{(1.1)^4} + \frac{35000}{(1.1)^5} \\
 &\quad + \frac{15000}{(1.1)^6} \\
 &= \text{Rs. } 15,808.13
 \end{aligned}$$

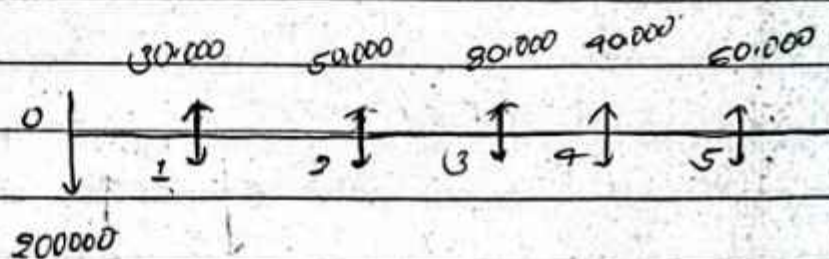
FM for uneven cashflow:

Evaluate the following project under odd FV Formulation

Year	0	1	2	3	4	5
Cash Flow	(200000)	30000	50000	80000	40000	60000

Interest = 10%

Cash Flow:



$$\begin{aligned}
 FV_{10\%} &= -200000 \times (1.1)^5 + 30000(1.1)^4 + 50000(1.1)^3 + 80000(1.1)^2 \\
 &\quad + (40000) \times (1.1)^1 + 60000 \\
 &= -RS. 10229
 \end{aligned}$$

Since, the calculated FM of the project is -ve, the project should be not selected.

→ Annual Worth Method (AW):

$$AW = \frac{-I}{PVIFA_{i,n} \text{ years}} + \frac{NAR + SV}{FVIFA_{i,n} \text{ years}}$$

Ex: → Evaluate the following project under AW formulation:

Initial Investment = 10,00,000

Annual Revenue = 1,50,000

Annual Cost = 20,000

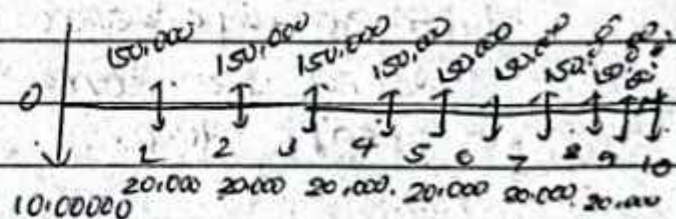
Salvage Value = 1,00,000

Useful life = 10 yrs ✓

1% MARR = 12% ✓

Cash Flow

Now, we have.



$$AW_{12\%} = \frac{-10,00,000}{5.6502} + \frac{1,30,000 + \frac{1,00,000}{17.5987}}{1}$$

$$= -91,286.45$$

Since, the calculated AW is -ve.

(Rate of Return Method)

② Rate of Return Method:

① ~~Rate of Return~~ IRR (Internal Rate of Return)

(It is compared with PW method)

⇒ IRR is the rate where $PW = 0$

⇒ It is the rate where present value of inflow and present value of outflow are equal

$$PW = -C + \frac{NAR}{(1+i)^n} = 0$$

Calculate IRR

Initial Investment: 10,00,000

Annual Revenue: 140,000

Annual Cost: 5,000

Salvage Value: 50,000

Useful life: 10 yrs.

MADE: 10%

Hint

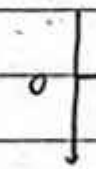
Factor = $\frac{\text{Initial Investment}}{\text{Average Net Annual Cash Flow}}$

$$= \frac{10,00,000}{135,000 + 50,000}$$

$$= \frac{10,00,000}{185,000}$$

$$= 5.35$$

$$= 7.14$$



$$0 = -10,00,000 + 135,000 \times \frac{(1+i)^{10} - 1}{i(1+i)^{10}} + 50,000 \times \frac{1}{(1+i)^{10}}$$

$$\Rightarrow 10,00,000 = 135,000 \times \frac{(1+i)^{10} - 1}{i(1+i)^{10}} + 50,000 \times \frac{1}{(1+i)^{10}}$$

$$\Rightarrow 1 = 0.135 \times \frac{(1+i)^{10} - 1}{i(1+i)^{10}} + \frac{0.05}{(1+i)^{10}}$$

Try at 7%

$$PW_{7\%} = -1000000 + 135000 \times PVIFA_{7\%, 10yrs} + \frac{50000}{(1.07)^{10}}$$

$$= -1000000 + 135000 \times 7.0236 + \frac{50000}{(1.07)^{10}}$$

$$= -99279.79$$

$$= -26309.027$$

Try at 6%

$$PW_{6\%} = -1000000 + 135000 \times 7.3601 + \frac{50000}{(1.06)^{10}}$$

$$= +21531.99$$

Interpolating for IRR

$$IRR = LR + \left[\frac{PW \text{ of LR}}{PW \text{ of LR} - PW \text{ of HR}} \times \text{Diff} \right]$$

$$= 6\% + \left[\frac{+21531.99}{21531.99 + 26309.027} \times 1 \right]$$

$$= 6.45\%$$

b) IRR for uneven cash flow:

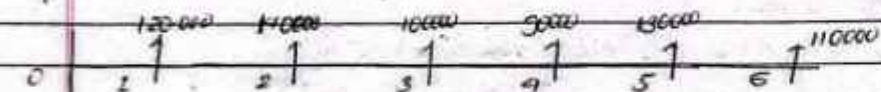
Evaluate the following under IRR formulation

Year	0	1	2	3	4	5	6
Cashflow	50000	12000	14000	10000	9000	13000	11000

MARR = 12%

Solution:

CFD



50000

Try at 10%

Form = II

NPV formula shown

$$\begin{aligned}
 PW_{10\%} &= -50000 + \frac{12000}{1.1} + \frac{14000}{(1.1)^2} + \frac{10000}{(1.1)^3} + \frac{9000}{(1.1)^4} + \frac{13000}{(1.1)^5} + \frac{11000}{(1.1)^6} \\
 &= -50000 + 10909.09 + 11348.54 + 7513.15 + 6195.12 + 8131.21 + 6195.12 \\
 &= +9207.96
 \end{aligned}$$

Try at 12%

$$\begin{aligned}
 PW_{12\%} &= -50000 + \frac{12000}{1.12} + \frac{14000}{(1.12)^2} + \frac{10000}{(1.12)^3} + \frac{9000}{(1.12)^4} + \frac{13000}{(1.12)^5} + \frac{11000}{(1.12)^6} \\
 &= -50000 + 10714.29 + 11058.21 + 7206.61 + 5917.17 + 7992.54 + 6000.00 \\
 &= -23060.93
 \end{aligned}$$

Interpolating for IRR.

$$IRR = LR + \left[\frac{PW \text{ of LR}}{PW \text{ of LR} - PW \text{ of HIR}} \times D\% \right]$$

$$= 10\% + \left[\frac{4207.96}{4207.96 + 23060.93} \times 2\% \right]$$

$$= 10\% + 0.305\%$$

$$= 10.305\%$$

Decision: If IRR is greater than given MARR then

select the project

if IRR is less than MARR then reject the project.

Here, calculated IRR of the project is less than MARR

so, this project is rejected.

IRR and unrecovered investment balance
Calculate IRR and also show up unrecovered balance
investments balance in table and graph

Initial Investment = 100000

Annual Revenue = 30,000

Annual Cost = 5000

Savage value = 5000

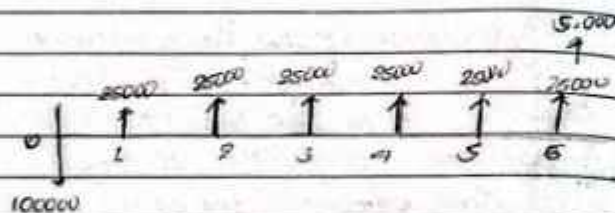
Interest Rate = 10%

Useful life = 6 yrs

Factor = $\frac{100000}{25000 + 5000}$

$\frac{25000 + 5000}{6}$

$\therefore 3.87$



Interest = 10%

Try at 14%

$$PW_{14\%} = -100000 + 25000 \times PAF(14\%, 6) + 5000$$

(2.195)

$$= +10483 = -509.56$$

Try at 13%

$$PW_{13\%} = -100000 + 25000 \times 3.9976 + 5000$$

(2.196)

$$= +2341.59$$

Interpolation for IRR, we have:

$$IRR = LR + \left[\frac{PW_{FLR} \times \Delta R}{PW_{FLR} - PW_{HLR}} \right]$$

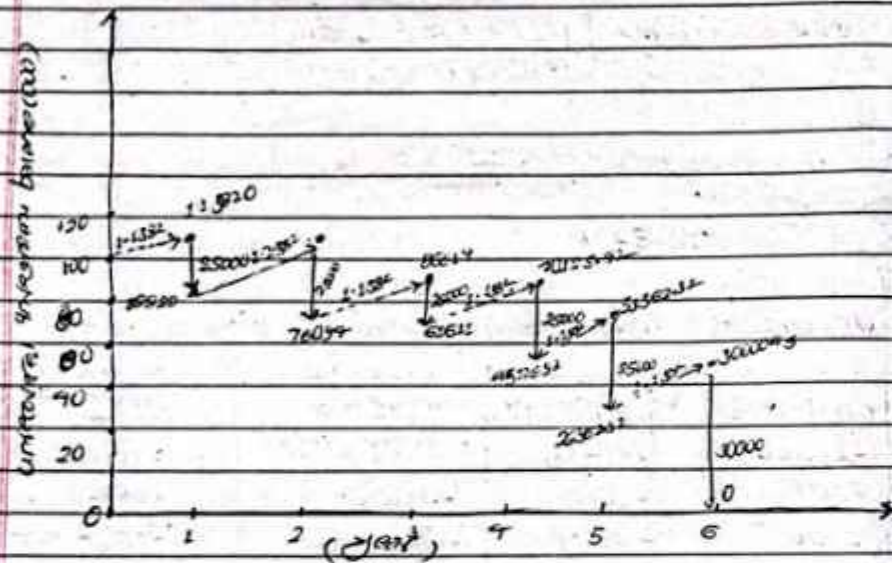
$$= 13\% + \left[\frac{2341.59 \times 1\%}{2341.59 - (-509.56)} \right]$$

$$= 13.82\%$$

unrecovered investment balance in tabular form

Year	Cash Flow	unrecovered investment at the beginning	(Interest) 2 Value at the end after deducting IRR	unrecovered investment at the end
0	-100000	-	-	-100000
1	25000	-100000	-113820	-88820
2	25000	-88820	-101099.979	-76099.979
3	25000	-76099.979	-86619.24	-61011.24
4	25000	-61011.24	-70125.92	-45125.92
5	25000	-45125.92	-51352.31	-26352.32
6	30000	-26352.32	-30000.95	0

Uncovered Investment balance in graph :-



Incremental IRR :-

Ex: Evaluate the following two projects under incremental IRR assumption.

Project A

Year	0	1	2	3	4	5
Cashflow	1000	100	120	130	140	1200

Project B

Year	0	1	2	3	4	5
Cashflow	2000	180	210	250	300	2400

MAARR = 10%

Solution :-

Calculation of Incremental cash flow.

Year	0	1	2	3	4	5
Cashflow	1000	80	90	120	160	1200

Calculation of IRR For Incremental cash flow :-

Try at 12%

$$PW_{12\%} = -1000 + \frac{80}{(1.12)^1} + \frac{90}{(1.12)^2} + \frac{120}{(1.12)^3} + \frac{160}{(1.12)^4} + \frac{1200}{(1.12)^5}$$

$$= 31.84$$

Try at 13%

$$PW_{13\%} = -1000 + \frac{80}{(1.13)^1} + \frac{90}{(1.13)^2} + \frac{120}{(1.13)^3} + \frac{160}{(1.13)^4} + \frac{1200}{(1.13)^5}$$

$$= -25.11$$

$$IRR = LR + \left[\frac{PW of LR}{PW of LR - PW of RR} \times Diff \right]$$

$$= 12\% + \left[\frac{21184}{111842511} \times 1 \right]$$

$$= 12.29\%$$

Decision Rules:-

Select the project with highest Investment if Incremental IRR is greater than given WACC.

Select the project with lower Investment if Incremental IRR is lower than given WACC.

Since in our case we select project B. Since calculated WACC is higher than given WACC.

⑥ ERR (External Rate of Return):

⇒ Following steps are used to calculate ERR

1) Discounting all cash outflow into zero year ✓

2) Compounding all cash inflow into final year ✓

3) ERR is calculated from the following way:-
Step 1 (1+ERR)ⁿ = Step 2

Ex:-

Consider the following cashflow of the project

$$IT = 25000$$

$$NAR = 8000$$

$$\text{Salvage value} = 5000$$

$$\text{Useful life} = 5 \text{ yr}$$

$$\text{WACC} = 20\%$$

Solution:-

Step 1

Discounting all cash outflow into zero year.

$$= 25000$$

CFD

Step 2:-

Compounding all cash inflow into 5th year.

$$= NAR \times FVIFA_{20\%, 5 \text{ yr}} + 5000$$

$$= 8000 \times 7.9916 + 5000$$

$$= 64532.8$$

negative

Step 3:- Establishing equivalence between step 1 & step 2

$$\text{Step 1} (1+ERR)^n = \text{Step 2}$$

$$\Rightarrow 25000 (1+ERR)^5 = 64532.8$$

$$\Rightarrow ERR = \frac{64532.8}{25000} \Rightarrow ERR = 0.2581$$

Since the calculated IRR is greater than given IRR.
So project is accepted.

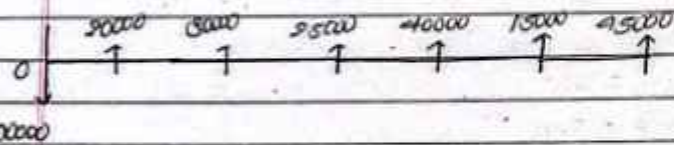
Here are given cash flow:

Evaluate the following project using IRR formulation

Year	0	1	2	3	4	5	6
Cash flow	-100000	20000	30000	25000	40000	15000	45000

IRR = 12%

Solution:- CF



Step 1: Discounting cash inflow and zero year
= 100000

Step 2: Compounding all cash inflow into the 5th year

$$= 20000 \times (1.12)^5 + 30000 \times (1.12)^4 + 25000 \times (1.12)^3 + 40000 \times (1.12)^2 + 15000 \times (1.12)^1 + 45000$$

$$= 229551.61$$

Step 3:

$$100000 \times (1 + IRR)^6 = 229551.61$$

$$\Rightarrow 107680 = 107680 \times 0.938$$

$$IRR =$$

A project has the following cash flows:

Year	0	1	2	3	4	5
Cash flow	-500	202	-2	196	3150	451

IRR = 10%

ERR = 14.14%

$\alpha = 2$

3. Payback period method (V.V.I)

$\Rightarrow$ (Time required to recover initial investment)

1. Simple payback period:

It does not follow the concept of time value of money.

Simple payback period = $\frac{\text{Initial Investment}}{\text{NAR (Net Annual Cash Flow)}}$

Example:-

$$II = \text{NAR} = 100000$$

$$\text{Annual Revenue} = 35000$$

$$\text{Annual cost} = 10000$$

$$\text{Salvage value} = 5000$$

$$\text{NAR} = 2000$$

$$\text{Life} = 10 \text{ yrs}$$

$$\text{Simple Payback Period} = \frac{100000}{35000 - 10000}$$

$$= \frac{100000}{25000} = 4 \text{ yrs}$$

Example of simple payback period for uneven

Calculate payback period.

Year	0	1	2	3	4	5	6
Cash flow	-10000	2000	3000	2500	4000	1500	9500

Year	Cash flow	Cumulative cash flow	
0	-10000	-10000	Payback period = 10000 year +
1	2000	-8000	Cumulative cash flow of
2	3000	-5000	10000 year
3	2500	-2500	Cash flow of right hand
4	4000	+1500	
5	1500	+3000	= 3 year + 2500
6	9500	+7500	4000

= 3.625 years

b) discounted payback period (sure)

⇒ consider time value of money (i.e. formula)

Calculate discounted payback period.

II = 10000

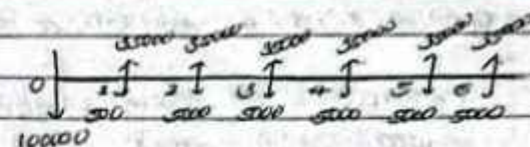
Annual revenue = 30000 (35000)

Annual cost = 5000

15% = 0.15

Salvage value = 10000

WACC = 10%



Calculation of discounted payback period

Year	Cash flow	Discounted cash flow	Cumulative discounted cash flow
0	-10000	-10000	-10000
1	3000	$30000/(1.1) = 27272.7$	-7272.7
2	3000	$3000/(1.1)^2 = 24793.38$	-4793.7
3	2500	$2500/(1.1)^3 = 22531.44$	-25395.26
4	3000	$3000/(1.1)^4 = 20090.90$	-9909.96
5	3000	$3000/(1.1)^5 = 18571.64$	+13722.78
6	4000	$4000/(1.1)^6 = 22578.95$	+36301.73

Interpolating for payback period

Payback period = 10000 year + $\frac{\text{Cumulative cash flow of } L_1}{\text{Discounted cash flow of } L_2}$

= 4 + 4795.8

18571.64

= 4.26704

Under cashflow and discount payback period.

Year	Cashflow	Discounted cashflow	Cumulative cashflow
0	-100000	-100000	-100000
1	40000	$40000/(1.1)^1 = 36363.6$	$-100000 + 36363.6 = -63636.4$
2	60000	$60000/(1.1)^2 = 49586.8$	$-63636.4 + 49586.8 = -14049.6$
3	30000	$30000/(1.1)^3 = 22539.8$	$-14049.6 + 22539.8 = 8490.2$
4	25000	$25000/(1.1)^4 = 17125.6$	$8490.2 + 17125.6 = 25615.8$

NPV = 10%

Payback period: $3 + \frac{14049.6}{25615.8}$

= 3.55 years

= 2 + $\frac{14049.6}{25615.8}$

= 2.64 years

Evaluate the following project under IRR formula.

Year	0	1	2	3	4	5	6	7	8
Cashflow	-1000000	200000	-100000	150000	180000	200000	200000	200000	200000

NPV = 0

Factor = $\frac{1000000}{200000}$

= 5

= 1120000 = 1120000

NPV = 0

$$PW_{IRR} = -1000000 + \frac{200000}{(1+IRR)^1} - \frac{100000}{(1+IRR)^2} + \frac{150000}{(1+IRR)^3} + \frac{180000}{(1+IRR)^4} + \frac{200000}{(1+IRR)^5} + \frac{200000}{(1+IRR)^6} + \frac{200000}{(1+IRR)^7} + \frac{200000}{(1+IRR)^8}$$

= -35793.92

IRR = 10%

$$PW_{IRR} = -1000000 + \frac{200000}{1.03} - \frac{100000}{(1.03)^2} + \frac{150000}{(1.03)^3} + \frac{180000}{(1.03)^4} + \frac{200000}{(1.03)^5} + \frac{200000}{(1.03)^6} + \frac{200000}{(1.03)^7} + \frac{200000}{(1.03)^8}$$

= 55596.14

Now, IRR formula

$$IRR = LR + \left[\frac{PW_{LR}}{PW_{LR} - PW_{LR+1}} \times (LR+1 - LR) \right]$$

$$= 3 + \left[\frac{55596.14}{55596.14 - 35793.92} \times 2 \right] = 9.20$$

97 B-C Ratio (Benefit-cost ratio) :- (V.V.I) 8 marks

$$\text{Benefit cost ratio} = \frac{\text{Total Benefit}}{\text{Total Cost}}$$

1) Conventional B-C Ratio

2) Modified BC Ratio

1) (a) PW Formulation :-

i) Initial Investment (PW)

ii) Annual Revenue (AW) (Benefit)

iii) Annual Cost (O & M)

iv) Salvage Value (FW)

$$\begin{aligned} \text{BC Ratio} &= \frac{\text{PW(B)} + \text{PW(S)}}{\text{PW(C)} + \text{PW(O \& M)}} \\ &= \frac{\text{PW(B)}}{\text{PW(C)} + \text{PW(O \& M)} - \text{PW(S)}} \end{aligned}$$

ii) FW Formulation :-

PW =

BC Ratio = FW(B)

$$\text{PW(C)} + \text{FW(O \& M)} - \text{FW(S)}$$

(c) AW Formulation :-

BC Ratio =

$$\frac{\text{AW(B)}}{\text{AW(C)} + \text{AW(O \& M)} - \text{AW(S)}}$$

2) Modified BC Ratio :-

i) PW Formulation :-

$$\begin{aligned} \text{BC} &= \frac{\text{PW(B)} + \text{PW(S)}}{\text{PW(C)} + \text{PW(O \& M)}} \end{aligned}$$

$$= \frac{\text{PW(B)} - \text{PW(O \& M)}}{\text{PW(C)} - \text{PW(S)}}$$

Example:

Calculate both types of BC ratio under PW/FW/AW

Formulation

$$I = 200000$$

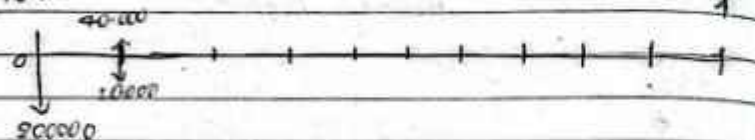
$$\text{Annual Revenue} = 40000$$

$$\text{Annual cost} = 10000$$

$$\text{Salvage value} = 35000$$

$$\text{useful life} = 10 \text{ yrs}$$

$$i = 10\%$$



conventional BC ratio under PW formulation:

we have

$$\text{BC ratio} = \frac{PW(B)}{PW(C)}$$

$$PW(C) = PW(0.5m) - PW(50)$$

* calculation of $PW(B)$

$$PW(B) = 40000 \times PVIFA_{10\%, 10 \text{ yrs}}$$

$$= 40000 \times 6.1446$$

$$= 245784$$

* calculation of $PW(C)$

$$PW(C) = 200000$$

* calculation of $PW(0.5m)$:

$$PW(0.5m) = 10000 \times PVIFA_{10\%, 10 \text{ yrs}}$$

$$= 10000 \times 6.1446$$

$$= 61446$$

* calculation of $PW(50)$

$$PW(50) = \frac{35000}{(1.1)^{10}}$$

$$= 18794.01$$

$$= 18794.01$$

$$\text{Now, BC ratio} = \frac{245784}{200000 + 61446 - 18794.01}$$

$$= 0.9911$$

$$= 0.9911$$

2) modified BC ratio:

$$\text{BC ratio} = \frac{PW(B) - PW(0.5m)}{PW(C) - PW(50)}$$

$$= \frac{245784 - 61446}{200000 - 18794.01}$$

$$= 245784 - 61446$$

$$= 200000 - 18794.01$$

$$= 0.98$$

conclusion:

Both type of method shows BC ratio < 1, so

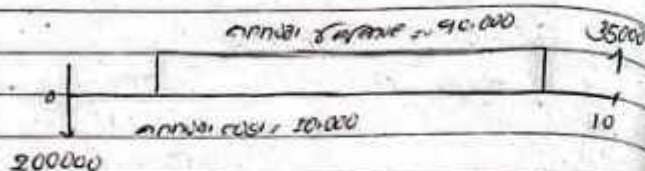
reject project

FW Formulation:

1. Conventional:

$$BC = FW(0)$$

$$FW(11) + FW(0.5m) - FW(50)$$



$$I = -200000$$

$$FW(0) = 90000$$

$$FW(0.5) = 3500$$

$$WAF = 10\%$$

$$NARR = 10\%$$

$$\begin{aligned} FW(0) &= 90,000 \times FVIFA_{10\%, 11} \\ &= 90,000 \times 15.9379 \\ &= 637,496 \end{aligned}$$

$$FW(11) = 200,000 \times (1.1)^{10} = 518,798.992$$

$$\begin{aligned} FW(0.5) &= 10,000 \times 15.9379 \\ &= 159,379 \end{aligned}$$

$$FW(50) = 35,000$$

$$BC = FW(0)$$

$$FW(11) + FW(0.5m) - FW(50)$$

$$= 637,496$$

$$518,798.992 + 159,379 - 35,000$$

$$= 0.99$$

2. Modified:

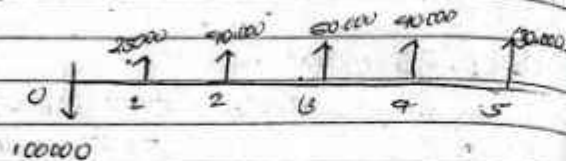
$$BC = 637,496 - 159,379 =$$

$$518,798.992 - 35,000$$

$$= 0.98$$

Evaluate the following project under both types of BC ratio with PW and FW Formulation

Year	0	1	2	3	4	5
Cash flow	-100000	25000	40000	60000	90000	30000



$$MARR = 10\%$$

$$S_0 = 0$$

$$Q_{SM} = 0$$

conventional BC ratio with pw form clear.

$$BC \text{ ratio} = \frac{PW(B)}{PW(C) + PW(OM) - PWS_0}$$

$$PW(B) = \frac{25000}{1.1} + \frac{40000}{(1.1)^2} + \frac{60000}{(1.1)^3} + \frac{90000}{(1.1)^4} + \frac{30000}{(1.1)^5}$$

$$= 146812.18$$

$$PW(C) = 100000$$

$$BC \text{ ratio} = \frac{146812.18}{100000} = 1.46$$

FW method:

$$FW(B) = 25000 \times (1.1)^4 + 40000 \times (1.1)^3 + 60000 \times (1.1)^2 + 90000 \times (1.1) + 30000$$

$$= 236942.5$$

$$FW(C) = 100000 \times (1.1)^5$$

$$= 161051$$

$$BC \text{ ratio} = \frac{FW(B)}{FW(C) + FW(OM) - FW(S_0)}$$

comparative analysis of alternatives :-

1) independent project cases :-

Suppose we have two independent projects A and B. We can consider the following combination :-

combination	Project A	Project B	Explanation
1	0	0	Do nothing
2	1	0	Select A
3	0	1	Select B
4	1	1	Select A and B

2) mutually exclusive project :-

Suppose we have three mutually exclusive projects A, B and C, then we can consider the following combination :-

combination	Project A	Project B	Project C	Explanation
1	0	0	0	Do nothing
2	1	0	0	Select A
3	0	1	0	Select B
4	0	0	1	Select C

3) contingent project case :-

Suppose, we have three projects A, B and C. Where A is contingent on the acceptance of both B and C, and C is contingent on acceptance of B, then we can consider following combination.

combination	Project A	Project B	Project C	Explanation
1	0	0	0	do nothing
2	1	1	1	Select A, B, C
3	0	1	0	Select B
4	0	1	1	Select B, C

Ex: government of Nepal has started four projects A, B, C, D for the expansion of IT in the country. The associated cashflow over 10 yrs are given in table below. The capital investment for each project is implied to 80,000 and MARR = 10% per year. Which investment alternative is best using FW formulation

Project:	A	B	C	D
Initial Invest.	32000	25000	72000	80000
Annual Revenue:	7000	5000	12000	16000
Salvage value:	4000	3000	5000	6000

A contingent on the acceptance of B

B contingent on the acceptance of C

Solution:-

Calculation of FW of all projects

FW of project A

$$\begin{aligned} FW_{10\%}(A) &= -32000 \times (1.1)^{-10} + 7000 \times [F/P 10\% \cdot 10\text{yr}] + 4000 \\ &= -32000 \times 0.377 + \\ &= -32000 \times (1.1)^{-10} + 7000 \times 5.9374 + 4000 \\ &= 32562.8 \end{aligned}$$

$$\begin{aligned} FW_{10\%}(B) &= -25000 \times (1.1)^{-10} + 5000 \times 5.9374 + 3000 \\ &= 17893.43 \end{aligned}$$

$$\begin{aligned} FW_{10\%}(C) &= -72000 \times (1.1)^{-10} + 12000 \times 5.9374 + 5000 \\ &= 9499.34 \end{aligned}$$

$$\begin{aligned} FW_{10\%}(D) &= -80000 \times (1.1)^{-10} + 16000 \times 5.9374 + 6000 \\ &= 18100.99 \\ &= 54399.03 \end{aligned}$$

Construction of the combination

Combination	Project A	Project B	Project C	Project D	Given
1	0	0	0	0	Do nothing
2	1	1	0	0	Accept A, B
3	0	1	0	0	Accept B
4	0	0	1	0	Accept C
5	0	0	1	1	Accept C, D

Combined project's investment and FW:-

Combination	Investment	FW
1	0	0
2	32k + 25k + 57000	50397.23
3	25000	17893.43
4	72000	9499.34
5	182000	63058.8

Let's take the combination

but if you find out the given that capital 80,000 we can choose 5

If the information capital is provided to 50,000 and the compensation is 9% referred, among the remaining consideration, compensation 2 is given to the highest FID.

Capitalize worth (CW)

$$CW = \frac{AW}{i + WARR}$$

$$AW = \frac{II}{P/F, i, n} + \frac{NAR + S}{P/F, i, n}$$

Compare structure M and N given below by the CW method, these structure are mutually exclusive.

ITEMS	STRUCTURE (M)	STRUCTURE (N)
Initial investment	12000	40,000
Salvage value	0	10,000
Annual cost	2200	1000
Useful life	10 yrs	25 yrs

$$WARR = 15\%$$

Solution:

We have,

$$CW = \frac{AW}{i + WARR}$$

Calculation of AW of structure 'M',

$$AW = \frac{12000 - 2200 + 0}{P/F, 15\%, 10}$$

$$= \frac{12000 - 2200}{5.0188}$$

$$= -4591.009$$

$$CW = \frac{-4591.009}{0.15}$$

$$= -30606.72$$

$$= -30606.72$$

Again, calculation of AW of structure 'N',

$$AW = \frac{40,000 - 1000 + 10,000}{P/F, 15\%, 25} + \frac{10,000}{P/F, 15\%, 25}$$

$$= \frac{40,000 - 1000 + 10,000}{6.4691} + \frac{10,000}{812.7930}$$

$$= -7191.02$$

$$NW: CW = \frac{-7191.02}{0.15}$$

$$= -47827.46$$

Both structures have -ve. C.I. structure 1st 4th
1st -ve same, we select structure 1st because
these structure are mutually exclusive.

Q.11

Repeatability:

Step 1: calculation of LCM of 14 & 18, i.e. of the project.

Step 2: calculation of PW of each project with Repeatability assumption.

Ex: → Evaluate the following two mutually exclusive projects under Repeatability assumption, with PW formulation.

Item	A	B
Initial investment	400000	600000
Annual Revenue	120000	160000
Annual Cost	10000	15000
Life	6 yrs	8 yrs
Salvage	40000	7000

Interest: 10%

Step 1: calculation of LCM of the life of 'A' and 'B'

2 | 6.8

3.4

2x3.4 = 6.8

Step 2: calculation of PW of the project 'A'

Project 'A' Repeats: 3.4 times

$$PW_{10\%}(A) = -400000 - 400000 - 400000 - 400000 + 110000 \times PVIFA_{10\%, 6.8} \\ + 40000 \times PVIF_{10\%, 6.8} \\ = (-400000 - 400000 + PVIFA_{10\%, 6.8} \times 110000 + PVIF_{10\%, 6.8} \times 40000)$$

Calculation of PW of the project 'B'

Project 'B' repeats 2.4 = 6 times

$$PW_{10\%}(B) = -600000 - 600000 - 600000 + 195000 \times PVIFA_{10\%, 6} \\ + 7000 \times PVIF_{10\%, 6}$$

3.4 CO-terminating

4 Evaluation of projects under co-terminal assumptions method

(1) first step: calculate FNI of lower life project

$$FNI = -II \times (1+i)^n + NAR \times FVF_{(1+i)^n, n, i} + SV$$

(2) for higher life project

(a) calculation of CR cost

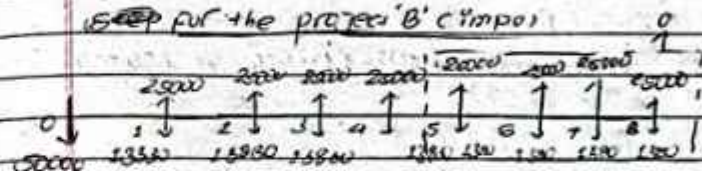
Ex. from the following information select the best project by using co-terminal assumption when interest is 10%.

Information	Project 'A'	Project 'B'
II	55000	50K
AR	16950	25K
AC	3K	13830
N	9 yrs	5 yrs

Step 1: calculation of FNI of the first project (Project A) (lower life).

$$\begin{aligned} FNI_{10\%, 10} &= -II \times (1+i)^n + NAR \times FVF_{(1+i)^n, n, i} + SV \\ &= -55000 \times (1.1)^9 + 16950 \times 5.9379 + 3000 + 0 \\ &= -11178 \end{aligned}$$

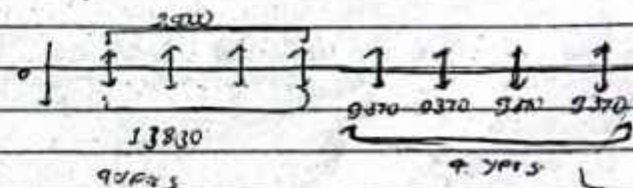
Step for the project B (higher)



In this case we can apply, in place of market value calculation approach (4 steps) (4mV)

Calculation of CR cost of the project B.

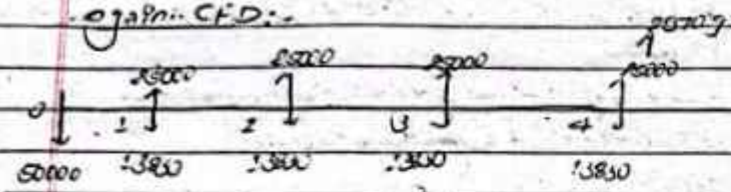
$$\begin{aligned} \text{CR cost} &= \frac{II}{N} + \frac{SV}{N} \\ &= \frac{50,000}{5} + \frac{0}{5} \\ &= 10,000 \end{aligned}$$



Calculation of CR cost present worth of C.R cost at the end of 9th year.

$$\begin{aligned} &= 10,000 \times FVF_{(1+i)^n, n, i} \\ &= 10,000 \times 5.9379 = 59,379 \end{aligned}$$

again CF D:-



Now calculation of Fw of Project B

$$FW_{\text{Project B}} = 5000 \times (1.1)^4 + 13170 \times \text{Factor} = 129709$$

$$= 8337$$

If these projects are mutually exclusive projects, the project A should be selected - & Project B should be rejected. Because project A has higher Fw than Project B (Fw of A > B).

These projects are mutually independent projects, both projects can be selected as both Fw is > 0

Chapter 1

Break even analysis

1. Break even analysis (CVP) :-

BEP $\Rightarrow$ no loss no gain

$$\text{Total Sales} = \text{Total Cost}$$

$$= \text{Fixed Cost} + \text{Variable Cost}$$

$$x \text{ units sales} = FC + (x \times VCPU)$$

Sales per unit Rs. 10

SP $\times$

$\rightarrow$ Fixed cost
depends on no. of units produced.

$\rightarrow$ Variable cost

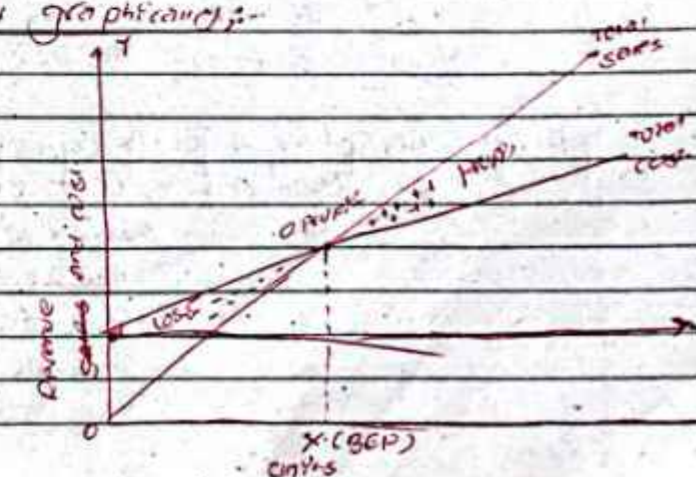
$\rightarrow$ depends on number of units produced

$\rightarrow$ different production to unit production

$$x = \frac{\text{Fixed Cost} + FC}{(SP - VC)}$$

$$BEP = \frac{FC}{SP - VC}$$

we can present it graphically:-



Suppose fixed cost associated with the production of products is Rs. 50,000. Assume that the variable cost is Rs. 20,000 and selling price is Rs. 30,000 for each product and minimum sales amount is Rs. 30,00,000 when sales volume is 100 units. Determine the break-even point and also plot it into BEP chart.

Solution:

Given:

Total fixed cost (Fc) = 50,000

Variable cost per unit (Vc) = 20,000

Selling price per unit (Sp) = 30,000

Number of units sold = 100 units

Total sales = 30,00,000

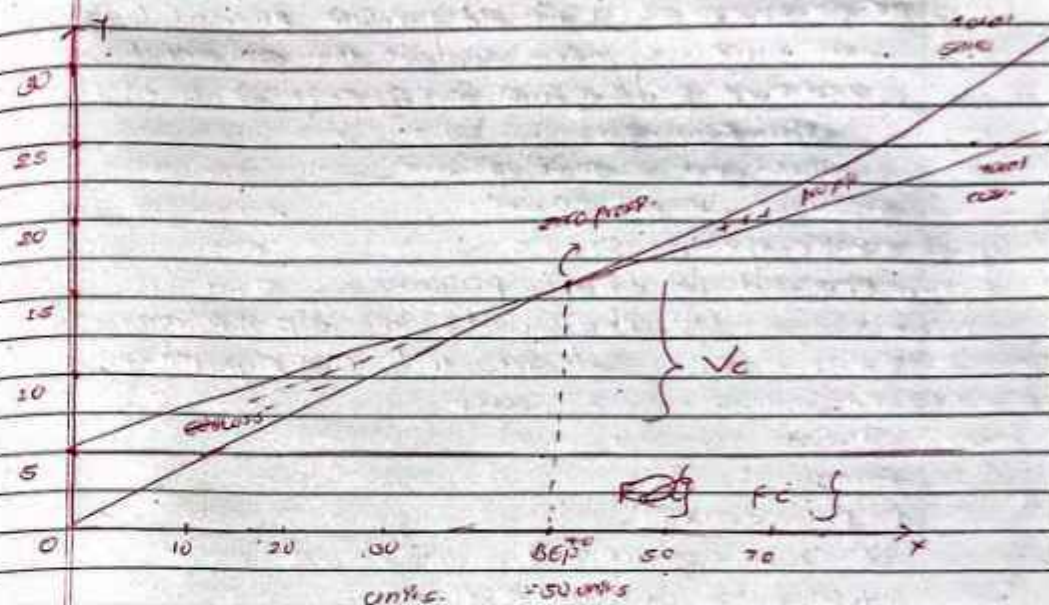
BEP = ?

We have:

$$\begin{aligned} \text{BEP} &= \frac{F_c}{S_p - V_c} \\ &= \frac{50,000}{30,000 - 20,000} \\ &= 50 \text{ units} \end{aligned}$$

For the BEP chart Let's calculate

$$\begin{aligned} \text{Total cost} &= F_c + \text{Total } V_c \\ &= 50,000 + 100 \times 20,000 \\ &= 20,50,000 \\ &= 20,50,000 \end{aligned}$$



(Produced) $\Rightarrow$ (many - times) $\Rightarrow$ out.

Q Sony company wants to produce Branded Laptops with increasing market demand. The company has estimated the following figures

$$F_c = 20,00,000$$

$$\text{Total cost} = 40,00,000$$

$$\text{Total sales} = 30,00,000$$

Sales quantity = 50 units

Required: (a) BE p of productions

(b) What should be the o/p if company desired to have the profit of 500000

Solution:

Given:

$$F_c = 20,00,000$$

$$\text{Total cost} = 40,00,000$$

$$\text{Total sales} = 30,00,000$$

Sales quantity = 50 units

$$V_c = \text{Total cost} - F_c$$

$$= 40,00,000 - 20,00,000$$

$$= 20,00,000$$

$$V_c(p) = \frac{20,00,000}{50} = 40,000$$

$$Sp(p) = \frac{30,00,000}{50} = 60,000$$

$$\therefore BGP = \frac{F_c}{Sp - V_c} = \frac{20,00,000}{60,000 - 40,000}$$

$$\frac{(p)}{(p)} = \frac{20,00,000}{20,000} = 100 \text{ units}$$

(6) Let's 'x' be the number of units sales to have 500000 profit.

$$500000 = \text{Total Sales} - \text{Total Cost}$$

$$= 500000 - 400000$$

$$500000 = (x + 60,000) - [FC + \text{Total Variable} (VC)]$$

$$500000 = 60,000x - [20,00000 + x \times 40,000]$$

⇒ ~~500~~

$$500000 = 60,000x - 20,00000 - 40,000x$$

$$\Rightarrow x = 125 \text{ units}$$

Break-even for dual projects:

Consider the following two motors each of 100 HP
oil capacity

124910m

	Motor A	Motor B
purchase cost	1.25K	1.60K
efficiency	74%	92%
life	20 yrs	10 yrs
maintenance cost	5000	2500
annual tax and insurance	1.5%	1.5%
MARR	15%	

Required: (a) How many hours per year would the motors have to be operated at full load for the annual cost to be equal if electricity cost is $\text{€ } 5/\text{kWh}$?

(b) If annual operation is more than 55 hours, which motor should be selected?

Solution:-

Calculation of annual equivalent cost of motor
 $= \frac{125000}{PVIFA_{15\%, 10}}$

1.7 Calculation of CR cost =

$$CR\ cost = \frac{125000}{PVIFA_{15\%, 10}} = \frac{125000}{5.0188} = 24906.35$$

2.7 maintenance cost = 5000

3.7 Annual tax and insurance = 1.5% of RS. 125000
 $= 1875$

4.7 Calculation of operating cost of power (electricity)

⇒ Let x be the number of operating per year.

$$Efficiency = \frac{Output}{Input}$$

$$Operating\ expenses = Input \times Rate \times hours$$

$$= \frac{Output \times Rate \times hours}{Efficiency}$$

$$= \frac{100 \times 0.746 \times 5 \times x}{0.74}$$

$$= 504.05x$$

[Note:- 1HP = 0.746 kW]

5. Total Annual Cost of motor = 1 + 2 + 3 + 4

$$= 24906.35 + 5000 + 1875 + 504.05x$$

$$= 31781.35 + 504.05x$$

For motor B

1) Calculation of annual equivalent cost

$$= \frac{160,000}{PVFA}$$

$$= \frac{160,000}{5.0188}$$

$$= 31880.13$$

2) Maintenance cost = Rs. 2500

3) Annual tax and insurance = 1.5% of 160,000

$$= \frac{160,000 \times 1.5}{100}$$

$$= 2400$$

4)

$$\text{Operating expenses} = \frac{200}{0.92} \times 0.796 \times 5 \times \alpha$$

$$= 405.43 \times$$

$$\therefore \text{Total Annual Cost of Motor 'B'} = 31880.13 + 2500 +$$

$$2400 + 405.43 \times$$

$$= 36780.13 + 405.43 \times$$

To get break-even point, we have,

$$\text{Annual equivalent cost of Motor A} = \text{Annual equivalent cost of Motor B}$$

$$\Rightarrow 31781.85 + 509.05 \alpha = 36780.13 + 405.43 \times$$

$$\Rightarrow 98.62 \alpha = 4998.28$$

$$\alpha = \frac{4998.28}{98.62} = 50.68 \text{ hrs} = 51 \text{ hrs}$$

$$98.62$$

Required (b) : If answer question 4 is more than

Note:- decision criteria:

If the anticipated level is below than Break even level, select the alternative with higher variable cost.

If the anticipated level is above than Break even level, select the alternative with lower variable cost.

Conclusion :- Motor 'B' is selected because of lower variable cost.

1. Diff?

(V.V.M)

(2) Sensitivity Analysis

Ques) Expt perform the sensitivity analysis for the following project over the range of $\pm 40\%$ in

- (a) II
- (b) Net Annual Revenue (NAR)
- (c) Salvage Value (SV)
- (d) Useful Life

II = Rs. 11500

NAR = Rs 3000

SV = 1000

Useful life (n) = 6 yr

MARR = 10%

Also draw a sensitivity graph and find most sensitivity and least sensitivity parameter.

Solution:-

C.F.D

-40%

0%

40%

* Calculation of PW of the given project (0%) (base)

$$\begin{aligned} PW_{10\%} &= -II + NAR \times PW_{10\%} \text{ factor} + SV / (1+i)^n \\ &= -11500 + 3000 \times 4.3553 + \frac{1000}{(1.1)^6} \end{aligned}$$

$$= 2130.8$$

* (A) when initial values $\pm 90\%$

$$i) +40\% \text{ } PW(10\%) = (-11500 \times 1.4) + 3000 \times 9.3553 + \frac{1000}{1.7715}$$

$$= -2470$$

$$ii) -90\% \text{ } PW(10\%) = (-11500 \times 0.6) + 13065.91 + 569.99$$

$$= 6730$$

B) when Net Annual Revenue varies $\pm 90\%$

$$i) +40\% \text{ } PW(10\%) = -11500 + (3000 \times 1.4) \times 9.3553 + 569.99$$

$$= 7356$$

$$ii) -90\% \text{ } PW(10\%) = -11500 + (3000 \times 0.6) \times 9.3553 + 569.99$$

$$= -3096$$

C) when Salvage value varies $\pm 90\%$

$$i) +40\% \text{ } , \text{ } PW(10\%) = -11500 + 3000 \times 9.3553 + \frac{1000 \times (1.4)}{1.7715}$$

$$= 2856$$

$$ii) -90\% \text{ } , \text{ } PW(10\%) = -11500 + 13065.91 + \frac{1000 \times 0.6}{1.7715}$$

$$= -2904$$

(D) when useful life varies $\pm 90\%$

$$UL = 6 \text{ yrs}$$

$$+40\% \rightarrow 6 \times 1.4 = 8.4 \text{ yrs}$$

$$-90\% \rightarrow 6 \times 0.6 = 3.6 \text{ yrs}$$

$$(i) \text{ 40\% } PWD(20) = -11500 + 3000 \times \frac{(1.1)^{20} - 1}{0.1(1.1)^{20}} + \frac{1000}{(1.1)^{20}}$$

$$= -11500 + 3000 \times \frac{(1.1)^{8.4} - 1}{(0.1)(1.1)^{8.4}} + \frac{1000}{(1.1)^{8.4}}$$

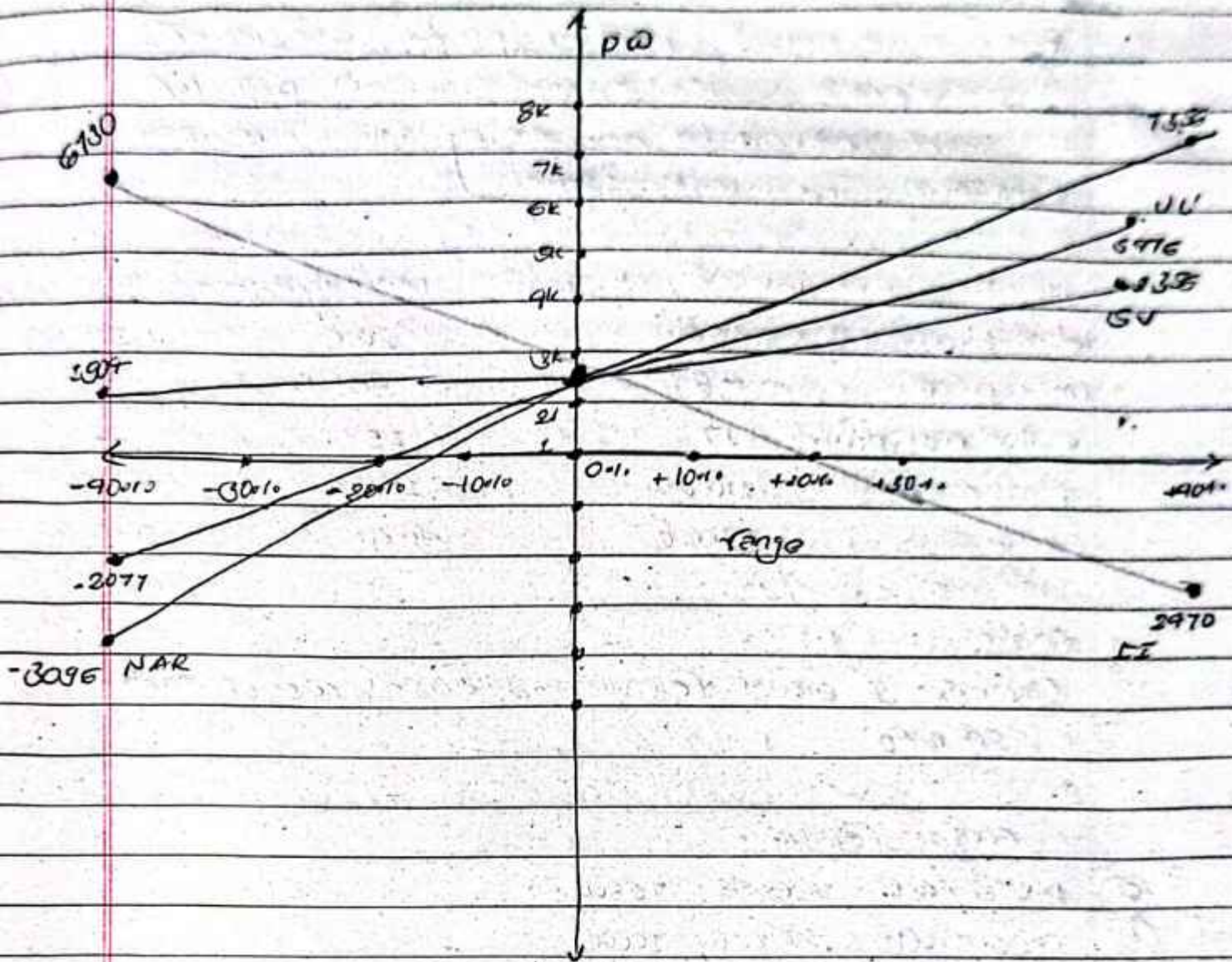
$$= 5476$$

$$(ii) \text{ 90\% } = -11500 + 3000 \times \frac{(1.1)^{3.6} - 1}{0.1(1.1)^{3.6}} + \frac{1000}{(1.1)^{3.6}}$$

$$= -2077$$

Calculation table for graph

Parameters	- 10%	0%	40%
(a) Initial Investment	6.730	2.130	- 9970
(b) Net annual Revenue	- 3096	2.130	7356
(c) Salvage value	1000	2.130	2356
(d) Useful life	- 2077	2.130	5476



On the basis of calculation of the table and sensitivity graph annual Revenue parameter project is highly sensitive when compare to other parameter because AR curve (i.e. Net Annual Revenue) curve is steeper than all other parameters in the graph. Similarly, savings value parameter is least sensitive parameter of the project.

8) Scenario analysis:-

From the following info of different variables of parameters and cases calculate p.u. for each scenario by assuming initial investment 125000 and $WARR = 15\%$.

Items	Worst case scenario	Most likely case	Best case
demand unit	1600	2000	2400
unit price	48	50	53
variable cost p unit	17	15	12
fixed cost	11000	10000	8000
Salvage value	30000	40000	50000

Solution:-

Calculation of annual revenue and annual cost of each scenario

Annual revenue

- (a) Worst case = $1600 \times 48 = 76800$
- (b) Most likely = $2000 \times 50 = 100000$
- (c) Best case = $2400 \times 53 = 127200$

Annual cost (FC + VC):-

- (a) Worst case $\Rightarrow 11000 + (1600 \times 17) = 38200$
- (b) Most likely = $10000 + (2000 \times 15) = 40000$
- (c) Best case $\Rightarrow 8000 + (2400 \times 12) = 36800$

Calculation of net present worth of WUGS case:

$$(A) \quad PW_{15\%} = -125000 + 76800 \times PVFA_{15\%, 5\%} + 30000$$

(1.15)⁵

$$= -125000 + (76800 - 38200) \times PVFA_{15\%, 5\%} + 30000$$

(1.15)⁵

(B) WUGS 18% case

$$PW_{18\%} = -125000 + (100000 - 40000) \times PVFA_{18\%, 5\%} + 10000$$

(1.18)⁵

(C) WUGS case:

$$PW_{25\%} = -125000 + (27200 - 36800) \times PVFA_{25\%} + 50000$$

(1.25)⁵

4) Decision tree

- ↳ It is a graphical presentation of alternatives
- ↳ It is also called tree diagram

Example: A glass factory that specializes in crystal is developing a substantial backlog and for this, the firm's management is considering 3 courses of actions. The current choice depends to largely upon the future demand, which may be low, medium and high. Show this decision situation in the form of decision tree and indicate most prefer decision and its corresponding expected monetary value (EMV)

Demand	Probability	S_1	S_2	S_3
Low	0.1	10	-20	-150
medium	0.5	50	60	20
High	0.4	50	100	200

where,

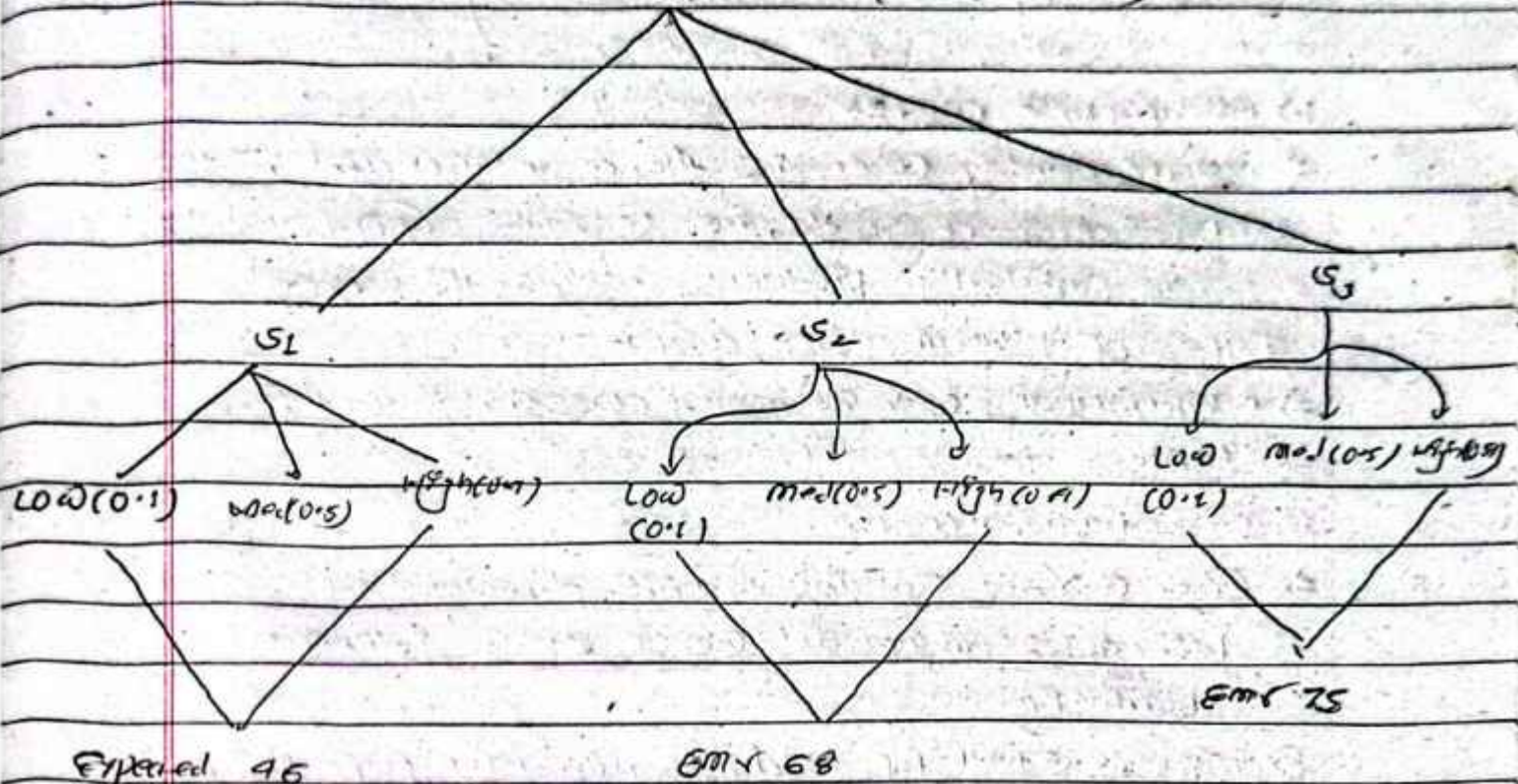
calculation of expected monetary value (EMV)

$$EMV(S_1) = 0.1 \times 10 + 0.5 \times 50 + 0.4 \times 50 = 48$$

$$EMV(S_2) = 0.1 \times (-20) + 0.5 \times 60 + 0.4 \times 100 = 68$$

$$EMV(S_3) = 0.1 \times (-150) + 0.5 \times 20 + 0.4 \times 200 = 75$$

Alternatives (choose option)



Decision value

Manufacturing and non-manufacturing cost

1. Manufacturing cost:

⇒ Manufacturing cost is the cost incurred in the production of goods, i.e. converting raw materials into finished products inside the factory.

Types of manufacturing cost:

⇒ Manufacturing cost is mainly classified into three types:

1. Direct material cost:

⇒ Cost of raw materials that are directly used in producing a product and can be easily identified with it.

Example: Steel for machines, wood for furniture, cotton for cloth.

2. Direct labour cost:

⇒ Wages paid to workers who are "directly involved" in the manufacturing process.

Example: Machine operators, assembly workers.

3. Manufacturing overhead (Factory overhead):

⇒ Indirect costs related to production that "cannot be" directly traced to a single product.

Examples:

- factory rent ✓
- electricity and water ✓
- machine depreciation ✓
- supervisor salary ✓
- factory maintenance ✓

2. Non-manufacturing cost:

⇒ Non-manufacturing cost is the cost incurred outside the production process, mainly for administration, selling and distribution of products.

Types of non-manufacturing cost:

1. Administrative cost:

⇒ Expenses related to managing and office operations.

Examples:

- office staff salary ✓
- office rent ✓
- stationery ✓
- legal and audit fees ✓

2. Selling cost:

⇒ Costs incurred to promote and sell products.

Examples:

- advertising ✓
- salesman commission ✓
- sales department salaries ✓
- showroom expenses ✓

3. Distribution cost:

⇒ Expenses related to delivering goods to customers.

Examples:

- Transportation cost ✓
- packing cost ✓
- warehouse charges ✓
- Delivery expenses ✓

1) Marginal cost (MC) :-

⇒ Marginal cost is the additional cost incurred by producing one more unit of good or service.

Definition :-

⇒ Marginal cost is the change in total cost when output is increased by one unit.

Formula :-

$$MC = \frac{\Delta TC}{\Delta Q}$$

Example :-

If the total cost of producing 10 units is Rs. 5,000

and producing 11 units costs Rs. 5,400,

then,

$$MC = 5,400 - 5,000 = \text{Rs. } 400$$

2) Opportunity cost :-

⇒ Opportunity cost is the value of the next best alternative foregone when one choice is made instead of another.

Definition :-

⇒ The benefit sacrificed by choosing one option over the next best alternative.

Example :-

If benefit sacrificed by choosing one option over the

if a student chooses to study engineering instead

of taking a job that pays Rs. 20,000 per month.

3. Sunk cost :-

⇒ Sunk cost is a cost that has already been incurred and cannot be recovered, regardless of future decisions.

Definition :-

⇒ Past cost that cannot be changed or avoided and should not affect current decision making.

Example :-

You paid Rs. 50,000 as a "non-refundable college admission fee". Even if you change colleges, the money is lost.

So, Rs. 50,000 is a "sunk cost".

prime cost = direct cost
overhead cost = indirect cost

(2 times) = 200 hours

Q. Following are the cost data for the production of 100 Barometer vacuums

Labor rate	Rs 40/hr
Lamph	50m @ Rs 200/m
Glu	300m @ Rs 50/m
Graphite	100 @ Rs 200
Labor hours	200 hrs
Total annual factory overhead	Rs 500000
Total annual direct labor hours	25000 hrs

Breakdown the costs into components of prime cost and overhead costs and find out the manufacturing cost of each vacuum.

Ans:

Calculation of total prime cost for 100 vacuums.

- (i) Labor = $50 \times 200 = 10000$
 - (ii) Glu = $300 \times 50 = 15000$
 - (iii) Graphite = $100 \times 200 = 20000$
 - (iv) Labor cost = $40 \times 200 = 8000$
- Total prime cost = 53

Overhead cost is

Total annual factory overhead = Rs 500000

Total annual labor hours = 25000

Cost for one hour = $\frac{500000}{25000} = Rs 20 \text{ per hour}$

Hours needed to produce 100 vacuums = 200 hours

Ex: $200 \times 100 = 200 \times 100 = 40000$ Overhead cost for 100 vacuums.

$\therefore$ Total cost = prime cost + overhead cost
 $= 53000 + 4000 = Rs 57000$

Per unit vacuum cost = $\frac{57000}{100} = Rs 570$

Life cycle cost :-

- ↳ In engineering practice, the term life cycle cost is often encountered. Life cycle cost is the sum of all expenditures associated with the item or project during its entire service life. The life cycle begins with identification of the economic need and ends with retirement and disposable activities. It may also be expressed as the summation of acquisition, operation, maintenance and disposable costs.
- ↳ Life cycle cost consists of following three components:
- First cost or initial investment; capital cost
 - Operating and maintenance costs.
 - Disposal costs.

VAT (Value Added Tax) :-

⇒ Some important points to be noted regarding VAT:

- It is an indirect tax.
- Consumers pay VAT to the businessman and every month business pays it to the government at the inland revenue office.
- Different countries have different rates.
- VAT is charged 13% and 15% in Nepal. Government does not charge VAT in some products like medicines, food, vegetables, meat etc.
- VAT is related with VAT Act 2052.
- It was first implemented in Nepal in 1990, 2059 B.S.
- VAT is a indirect sales tax, an expenditure tax.

- Businesses can claim credit for VAT paid on purchases (input tax) against VAT collected on sales (output tax).
- Like entry or payment, main reason for fines and interest charges.
- Businesses with an annual turnover exceeding NPR 5 million must register for VAT.
- Generally, the 25th day of the following month is the deadline of paying tax.

Evaluate the following project under IRR

Formulation:

$$\text{Initial} = 500000$$

$$\text{AE} = 100000$$

$$\text{AC} = 20000$$

$$\text{SV} = 50000$$

$$\text{Interest} = 10\%$$

$$\text{Useful} = 10 \text{ yrs.}$$

$$\text{Factor} = \frac{500000}{50000 + 80000}$$

$$= \frac{50000 + 80000}{10}$$

$$= 6211$$

Solution:

Try 10%:

$$\begin{aligned} \text{PW}_{10\%} &= -500000 + 80000 + \text{PW of } 100000 \times 10 \text{ yrs} + \frac{50000}{(1.1)^{10}} \\ &= 132623 \end{aligned}$$

Try 20%:

$$\begin{aligned} \text{PW}_{20\%} &= -500000 + 80000 + \text{PW of } 100000 \times 10 \text{ yrs} + \frac{50000}{(1.2)^{10}} \\ &= 10895 \end{aligned}$$

Try 25%:

$$\begin{aligned} \text{PW}_{25\%} &= -500000 + 80000 + \text{PW of } 100000 \times 10 \text{ yrs} + \frac{50000}{(1.25)^{10}} \\ &= -9586 \end{aligned}$$

Now interpolating IRR:

$$\begin{aligned} \text{IRR} &= \text{LR} + \left[\frac{\text{PW of LR} \times \text{diff.}}{\text{PW of LR} - \text{PW of HR}} \right] \\ &= 10 + \left[\frac{10895}{10895 + 9586} \right] \times 15 = 10.53 \approx 11\% \end{aligned}$$

Since, the calculated IRR of the project is 11% which is greater than given MARR, so accept project.

② SOLUTIONS

Constraint to be formed = 5 FROM 8 boys and 4 girls

① EXACTLY 2 girls

(i) at least 2 girls

(ii) at most 2 girls

① EXACTLY 2 girls OUT OF 5 members, remaining boy $(5-2)=3$
 $4C_2 \times 8C_3$
 $= 6 \times 56 \text{ ways}$
 $= 336 \text{ ways}$

② At least 2 girls (≥ 2 , 3, 4 girls)

① $4C_2 \times 8C_3 = 6 \times 56 = 336$

② $4C_3 \times 8C_2 = 4 \times 28 = 112$

③ $4C_4 \times 8C_1 = 1 \times 8 = 8$

$\therefore \text{TOTAL} = 336 + 112 + 8 = 456$

81 ways, 52
 567, 8,
 5
 56
 5
 56

③ At most 2 girls (≤ 2 , 0 girls, 1 girl, 2 girls)

① 0 girls (all boys)

$8C_5 = 56$

② 1 girl

$4C_1 \times 8C_4 = 4 \times 70 = 280$

③ 2 girls

$4C_2 \times 8C_3 = 6 \times 56 = 336$

$\therefore \text{TOTAL} = 56 + 280 + 336 = 672$

⑤ The annual fuel cost required to operate

SOLUTION

INITIAL FUEL PRICE (Y1) = 61

INFLATION RATE (I) = 5%

Year	Cash flow of fuel	Conversion factor (CF)	Present value
1			
2			
3			
4			
5			

The initial fuel price will be

$2.4 = (1.05)^n (1.05)$

$9 = (1.05 \times 1.05) - 1$

$= 5\% \text{ per year}$

The present equivalent cost will be

• PW of actual cost (12.3) = $\frac{210000}{1.113} + \frac{220000}{1.113^2} + \frac{230000}{1.113^3}$

$= \frac{213100}{1.113} = 673020$

$$I = 20,000$$

$$I_{\text{rate}} = 5000 (1.5\% \text{ per year})$$

$$I_{\text{rate}} = 5000 (1.5\% \text{ per year})$$

$$\text{Now, } P_0(S_1) = 5000(1.10 \cdot 0.5)^{-1} + 5380(1.05)^{-2} + 5.775(1.05)^{-3}$$

$$= 29,725$$

Since, the value of P_0 is 20,000 and not the value of P_0 .

Cost of Capital

$$K_d(1) = K_d(1-t)$$

$$K_d(1) = \left(\frac{I}{NP} \right) \times (1-t)$$

Initial I = Interest Amount

NP = Net proceeds or price of the bond or determine

K_d = Cost of debt or interest rate

t = tax rate

NP = Net proceeds

NP = Value of bond - flotation cost

I = amount of interest

we have,

$$K_d(1) = K_d(1-t)$$

$$= 12\% (1-0.35)$$

cost of debt / bond

Q. Given,

Value of bond: Rs. 1000

Flotation cost: 3% of 3% of 1000 = 30

$$12\% = 10.1\%$$

$$K_d = 10.1\%$$

$$K_d = \frac{I}{NP} (1 - t)$$

Interest (I) = 10% of 1000 = Rs. 100

Net proceeds = V₀ - Flotation cost

$$= 1000 - 30$$

$$= 970$$

$$= \left(\frac{100 \times 100 (1 - 0.1)}{970} \right)$$

when non-hybrid maturity period (i.e. n) is non-perpetual

$$= \left(\frac{I + \frac{M - V}{n}}{NP} \right) (1 - t)$$

M = maturity value

V = market value

Notes: Selling price of the bond on market is market value (V)

Interest (I) is calculated on the basis of maturity value.

Cost of preferred stock and cost of bond are calculated in same way except following things are different

- Interest on preferred dividend is intended i.e. (D)
- (1 - t) is not in preferred stock (because)

cost of common stock

$$K_e = \frac{D_1}{NP} + g$$

where, K_e = cost of common stock or capital

D₁ = Expected dividend this year

g = growth rate

D₀ = last year dividend

$$D_1 = D_0 (1 + g)$$

Q.

Given,

Expected dividend D₁ = Rs. 12

Growth rate g =

Inflation:

⇒ Inflation is the continuous rise in the general price level of goods and services over time, which reduces the purchasing power of money.

Measures of Inflation:

1) Consumer Price Index (CPI)

⇒ measures changes in prices of goods and services commonly used by consumers (Food, clothing, rent, transport).

⇒ most widely used measure of inflation.

2) Producer Price Index:

⇒ measures price changes received by producers for their goods.

⇒ shows inflation from the supply side.

3) Wholesale Price Index

⇒ measures price changes of goods at the wholesale level (before retail).

⇒ useful for tracking inflation in "producers" and "trade".

4) GDP Deflator:

⇒ measures price changes of all goods and services produced in an economy.

⇒ Broadest measure of inflation.

ACTUAL DOLLARS (NOMINAL DOLLARS):

- ⇒ Actual dollars refer to the value of money measured in the prices of the current year.
- ⇒ they include the effect of inflation.
- ⇒ they show money as it is, including past or present in that year.

Example:

- Salary in 2015 = RS. 20,000
- Salary in 2025 = RS. 40,000
- i.e., RS. 40,000 in 2025 is actual (nominal) dollars.

Constant Dollars (Real Dollars):

- ⇒ Constant dollars refer to the value of money after removing the effect of inflation.
- ⇒ they are measured using prices of a fixed base year.
- ⇒ they show the real purchasing power of money.

Example:

- Taking 2015 as the base year.
- RS. 40,000 in 2025 becomes RS. 20,000 in 2015 prices.
- RS. 20,000 is the constant (real) dollars.

Key difference:

- Actual dollars = value at current prices (with inflation)
- Constant dollars = value at fixed prices (without inflation)

Ex: The annual fuel costs required to operate a small solid waste treatment plant are projected to be RS. 20,000. Without considering any future inflation, the best estimate indicates that the annual inflation rate interest i will be 6% and the general inflation rate p will be 5%. If the plant has the facility, users will be for 4 years, what is the present equivalent of RS 6 fuel costs? Use actual dollar method.

Solution:

Inflation free rate of interest (i): 6%

Inflation rate (p): 5%

Year	Cash flow of fuel	Conversion factor (F) = $(1+i)^n (1-p)^n$	Actual cash flow
1	20000	$(1.06)^1 (1.05)^1$	21000
2	20000	$(1.06)^2 (1.05)^2$	22050
3	20000	$(1.06)^3 (1.05)^3$	23155
4	20000	$(1.06)^4 (1.05)^4$	24310

The market interest rate will be:

$$i = (1+i)^n (1+p)^n$$

$$i = (1+0.06) (1+0.05) - 1$$

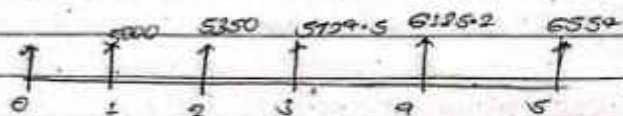
$$= 11.31\%$$

The present equivalent cost will be:

$$P_0 \text{ of actual cost } (i) = \frac{21000}{(1.1131)^1} + \frac{22050}{(1.1131)^2} + \frac{23155}{(1.1131)^3} + \frac{24310}{(1.1131)^4}$$

- ③ A series of five consumer dollar-for-dollar prizes (beginning with 5000 at the end of the first year) are increasing at the rate of 7% per year for five years. Ignoring the 7% rate is 5% and inflation is 4% feasible investment of investment cost is 20,000?

Q: What?



$$\begin{aligned}
 PV(CF) &= 5,000(1.05)^{-1} + 5,350(1.05)^{-2} + 5,711.5(1.05)^{-3} \\
 &\quad + 6,125.2(1.05)^{-4} + 6,559(1.05)^{-5} \\
 &= 24,725
 \end{aligned}$$

Since investment cost is 20,000 and equivalent net benefit is 24,725.

there is benefit to investment.

11 Methods of financing / sources of capital / sources of investment

⇒ Financing refers to the process of raising funds required for starting, operating, and expanding a business. The funds (capital) can be obtained from different sources, mainly through equity financing and debt financing.

1.7 Equity financing:

⇒ Equity financing is the ownership financing of a business. In this method, funds are raised by issuing shares to investors, making them owners of the company.

Major features of Equity financing:

- Interest is not paid in equity financing. ✓
- Investors are paid dividend, but the dividend is not fixed. ✓
- Dividend is paid only when the company earns profit. ✓
- Generally, there is no maturity period in equity financing capital, so it is a permanent source of finance. ✓
- Equity shareholders have ownership rights to vote in the company. ✓
- Equity financing is not concerned with any loan or debt, hence no interest payment. ✓

Types of Equity financing :-

1) Common (Equity) Stock financing :-

- ⇒ owning shares/stocks of the company
- ⇒ they have 0 voting rights
- ⇒ higher risk but higher return

2) Preferred Stock financing :-

- ⇒ shareholders receive dividend as a fixed rate
- ⇒ generally, no voting rights
- ⇒ preference over equity shareholders in dividend payment

3) Retained Earnings financing :-

- profits retained and reinvested in the business
- no cost involved
- no external borrowing required

Note: Debt is riskier than equity financing

2. Debt financing :-

- ⇒ debt financing refers to raising funds through borrowed capital that must be repaid after a certain period.

Major features of Debt financing :-

- Interest is paid on debt financing
- the interest rate is fixed
- Debt has a fixed maturity period
- Ownership of the company does not change
- Interest must be paid even if the company incurs loss

Sources of Debt financing :-

- ⇒ Bank loans
- ⇒ Debentures/Bonds
- ⇒ Loans from financial institutions

Cost of debt/bond :-

$$K_d(1) = K_d(1-t)$$

$$K_d(1) = \left(\frac{I}{NP} \right) \times (1-t)$$

I = Amount of interest

NP = Net proceeds

$K_d(1)$ = Cost of debt after tax

t = tax rate

NP = Value of bond - Flotation cost

Q. Garshad trading company financing plans for next year include the sale of long term bonds with a 10% coupon. The company believes it can sell the bonds at a price that will give a yield to maturity of 12 percent. If the tax rate is 35 percent, what is the company after tax cost of debt?

Solution

$$K_d(1) = K_d(1-t)$$
$$= 12(1-0.35)$$

Example 2: A company has issued Rs 1000 per value debenture with 3% flotation cost. The coupon rate of debenture is 10% and tax rate is 30%. Calculate the cost of this debt.

Solution:

Value of the debenture $V_d = \text{Rs. } 1000$

Flotation cost $FC = 3\% \text{ of Rs. } 1000 = \text{Rs. } 30$

Net proceeds $N_d = 1000 - 30 = 970$

Tax rate $(t) = 30\%$

Coupon rate $(r) = 10\% \text{ of } 1000 = \text{Rs. } 100$

We have,

$$1000 = 970 (1 - 0.3)$$

$$\therefore 6.10\%$$

Note: n is having maturity period (y.m.n) = non-perpetual

$$= \left(\frac{1 + \frac{1+r-V}{n}}{\frac{1+r+V}{3}} \right) (1-t)$$

r : maturity rate

V : market value

⇒ Selling price of the bond in market is market value denoted by V

⇒ Interest (I) is calculated on the basis of market value.

Cost of preferred stock and cost of bond are calculated in same way except following thing are:

- ① In cost of preferred stock instead of interest, dividend (D) is given.
- ② $(1-t)$ is not in preferred stock calculation.

Cost of common stock:

$$K_e = \frac{D_1}{NP} + g$$

K_e : Cost of common stock or Equity

D_1 : Expected dividend this year

g : growth rate

D_0 : Last year dividend

$$D_1 = D_0 (1+g)$$

Ex: A bank is making new issue of common stock of Rs. 100 each. It is expected dividend is 12 and expected to grow at the rate of 8 percent. The issue cost is 15 percent.

Solution:

Expected dividend $D_1 = \text{Rs. } 12$

Growth rate $g = 8\%$

Price of the stock $P_0 = \text{Rs. } 100$

Issue cost = 15%

We have,

$$\text{Cost of common equity } (K_e) = \frac{D_1}{NP} + g$$

$$= \frac{12}{100 \times 95} + 0.08$$

=

Example: A company issues 10 parval fire insurance preference shares. the face value per share is Rs. 100, but the issue price is Rs. 85. what is the cost of a parval stock? what is the issue price is Rs. 85. 100?

SUNFISH:

Gilman.

Fair Value of preferred stock = Rs. 100

initial value of prefinal score = 85

$$\text{Dividend yield} = 10\%$$

Dividend on the preferred stock (Dps) = 10% of Ds = 100

we have.

RS-10

$$kr: DP5/NP: \frac{10}{85} = 0.1176 \text{ or } 11.76\%$$

Again, if the resale price is Rs. 110.

$$\text{lep: } Dps / np = \frac{1.0}{110} = 0.10 \text{ or } 10\%$$

5. compute the component cost and calculate average cost of capital from the financial information extracted from Shan Yungwei, excerpta

Marginal Income Tax Rate: 20%

Common stock of Yppas 100 % National Yppas 93
Expected dividend and growth rate are

Ruppre's 100 preferred stock, dividend of 8% and
Pleasant Corp is 20%, net profit

Calculation of Component cost of each capital.

Cost of Debt

Given,

$$K_d = 12\%$$

$$\text{Tax} = 34\%$$

$$K_d(1) = 3$$

$$K_d(1) = K_d(1 - t)$$

$$= 12\% \times (1 - 0.34)$$

$$= 12\% \times 0.66$$

$$= 7.92\%$$

Cost of Common Stock

Given,

Value of common stock = ₹10

$$\text{New Price} = 98$$

$$\text{Dividend} = 3$$

$$\text{Growth rate} = 7\%$$

$$K_e = ?$$

$$K_e = \frac{D_1}{NP} + g$$

$$= \frac{3}{98} + 0.07$$

$$= 0.0306$$

$$= 10.25\%$$

Cost of Preferred Stock

Given

$$\text{Value} = 100$$

$$\text{Dividend} = 10\%$$

$$NP = (100 - 2) = 98$$

$$\text{Dividend} = 100 \times 10\% = 10$$

So here,

$$K_p = \frac{\text{Dividend}}{NP} \times 100$$

$$= \frac{10}{98} \times 100$$

$$= 10.204\%$$

$$= 10.20\%$$

Calculation of WACC :-

Capital	Amount	Weight	Cost of Capital	Weight x Cost
Debt	400	0.4	7.92%	
Common stock	400	0.4	10.25%	
Preferred stock	200	0.2	10.20%	
	1000	1		Total = 9.60%

Calculation of Economic Service Life (ESL)

- ① The economic service life of an asset is defined as the period of useful life that minimizes the annual equivalent cost (AEC) of operating the asset.

$$AEC = CR + OC$$

where,

CR = Capital Recovery cost

OC = ^{Annual} operating cost

Ex: Suppose

- ② Suppose a company decided to have a new machine of cost ₹ 1,80,000 and useful life is 18 years. The operation cost ₹ 5,300 in the 1st year and the cost per year increases by 15% ^{per} year. Salvage value is declined each year by 2% from the previous year salvage value. Find the economic service life, where

$$MARR = 12\%$$

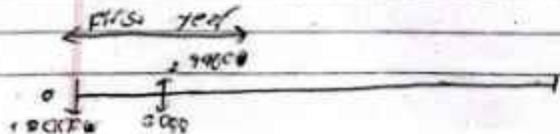
Sol: Given

$$\text{Salvage value} = (\text{0 year investment}) \times 0.8 \quad (SV_1)$$

Calculation of AEC for every year.

We have,

$$AEC = CR + OC$$



$$\text{Salvage value} = 1,80,000 \times 0.8 = 1,44,000$$

Calculation of CR cost.

$$= \frac{1,80,000}{PVIFA_{12\%, 18\text{yr}}} - \frac{1,44,000}{FVIFA_{12\%, 18\text{yr}}}$$

$$= \frac{1,80,000}{1} - \frac{1,44,000}{1}$$

$$= 5760$$

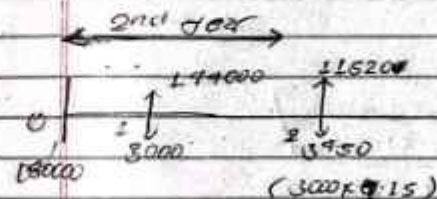
Calculation of OC:

$$OC = \frac{5000}{1.12}$$

$$PVIFA_{12\%, 18\text{yr}}$$

$$OC = 3000$$

$$AEC \text{ of 1st year} = 5760 + 3000 = 8760$$



$$\begin{aligned} CR \text{ cost} &= \frac{1,80,000}{PVIFA_{12\%, 17\text{yr}}} - \frac{1,52,000}{FVIFA_{12\%, 17\text{yr}}} \\ &= 5917 \end{aligned}$$

Calculation of OC:

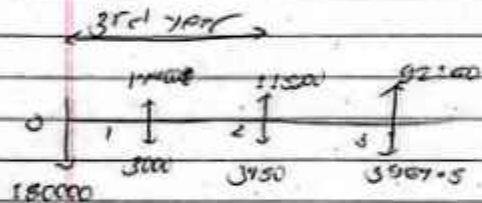
$$OC = \frac{3000 + \frac{3950}{(1.12)^1}}{1.12}$$

PVIFA 12%, 2 yrs

$$= 3212$$

$$ABC \text{ of 2nd year} = CR + OC$$

$$= 5214 + 3212 = 8426$$



$$CR \text{ cost} = 18000 - 0.210$$

PVIFA 12%, 3 yrs FVIFA 12%, 3 yrs

$$= 1763$$

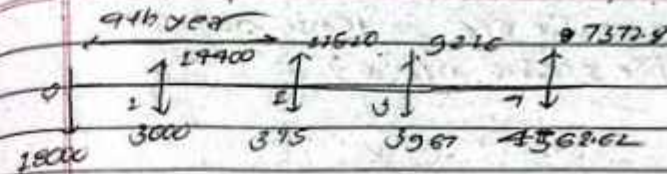
Calculation of OC :-

$$OC = \frac{\left[\frac{3000}{1.12} + \frac{3950}{(1.12)^2} + \frac{3967.5}{(1.12)^3} \right]}{PVIFA 12\%, 3 \text{ yrs}}$$

$$= 3936$$

$$ABC = 1763 + 3936$$

$$= 5699$$



$$CR \text{ cost} = 18000 - 7572.6$$

PVIFA 12%, 4 yrs FVIFA 12%, 4 yrs

Calculation of OC:

$$OC = \frac{\left[\frac{3000}{1.12} + \frac{3950}{(1.12)^2} + \frac{3967.5}{(1.12)^3} + \frac{4562.6}{(1.12)^4} \right]}{PVIFA 12\%, 4 \text{ yrs}}$$

$$ABC = 8055$$

5th
ABC

Calculation of ABC cost in 5th year

Conclusion, 5th year we have 7077 ABC cost and 4% started to increase in 5th year, so we didn't have to


 Date _____
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1st charge = 4000
cost = 4000

IRREVERSIBEL
porbac.

60-70%

• Repetition

Sensitivity

41000

INPARKIN

depreciation

(CNS-1406A)

(pH- method) $\Rightarrow$ best method.

$$P_{12}(1.04) = 5,15000 \times p_{12}(1.04, 2) + 5,85000 \times p_{12}(1.04, 9) + 5,86000 (p_{12}(1.04, 5) + p_{12}(1.04, 7))$$

30153, 102-5.